Onboarding Databases to DSF Hub Reference Guide

25 June 2026

Contents

Google Cloud (GCP) Data Sources

DSF Hub supports the following Google Cloud Platform (GCP) data sources. Please select a link below to view the step-by-step instructions for each specific platform.

- AlloyDB for PostgreSQL Onboarding Steps
- BigQuery Onboarding Steps
- Bigtable Onboarding Steps
- Cloud SQL for MySQL Onboarding Steps
- Cloud SQL for PostgreSQL Onboarding Steps
  - Cloud SQL for PostgreSQL Performance Improvements for v 4.16+
- Cloud SQL for SQL Server Data Sources
  - Cloud SQL for SQL Server with PubSub Onboarding Steps
  - Cloud SQL for SQL Server with Storage Bucket Onboarding Steps
- Firestore Onboarding Steps
- Spanner Onboarding Steps

Document generated by Confluence on Jun 19, 2026 01:21

Atlassian

AlloyDB for PostgreSQL Onboarding Steps

This topic reviews the steps required to audit activity on AlloyDB for PostgreSQL databases and send that information to DSF Hub for auditing, analysis, detecting and preventing security events.

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The following main topics are included:

- An overview of the available scope for the data source, given its native audit data.
- A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.
- Instructions on how to enable audit on the data source and collect data using DSF Hub.
- Initial troubleshooting steps and technical support information.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps. Please click on each of the steps below to display the content.

STEP 1: Onboarding Prerequisites

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

Permissions for Google Cloud Account Principal

Grant the following IAM roles to the Google cloud account principal:

- **Logging Admin** and **Project IAM Admin** roles: for enabling Data Access logs, viewing logs and creating a Log Router Sink.
- **Pub/Sub Admin** role: for creating a Pub/Sub Topic and Subscription.
- **AlloyDB Admin** role: for modifying database-specific flags.
- **Service Account Admin** and optionally **Service Account Key Admin** roles: for creating a service account and generating keys (if using the "service_account" authentication mechanism). See Permissions and roles for more information on the list of individual permissions contained in each predefined role.

Creating a Google Cloud Service Account

DSF supports the following two authentication mechanisms for accessing Google Cloud resources. Note that both authentication mechanisms make use of a Service Account. The process for creating a Service Account is described in Creating and managing service accounts.

- **Service Account:** to use the `service_account` authentication mechanism, a JSON-formatted key file needs to be generated holding the credentials that the Agentless Gateway will use to access audit logs from the Pub/Sub subscription. Ensure that the key file is copied to the Agentless Gateway host and it is owned by the `sonarw` user. For more information on how to create a service account key, see Creating and managing service account keys.

- **Default:** to use the `default` authentication mechanism, attach the Service Account created above to the Agentless Gateway hosted on a Google Cloud Compute Engine virtual machine instance. For more information on how to attach a Service Account to a GCP Compute Engine virtual machine, see Change the attached service account.

The Service Account should be granted the following roles:

- Pub/Sub Subscriber
- Pub/Sub Viewer
- Viewer

> **Note:**
>
> It is recommended to keep both the Service Account and the Agentless Gateway Google Cloud VM instance within the same Google Cloud Project.

Google Cloud Log Routing Overview: Log Router Sinks, Pub/Sub Topics and Subscriptions

Log Router Sinks control how Logging routes logs in Google Cloud. Using sinks, you can route audit logs to one or more supported destinations. Please follow the instructions in Route logs to supported destinations. The following summarizes the steps to be completed:

1. Create a Log Router sink that sends the audit logs to a new or existing Pub/Sub Topic as the destination service.
    a. In the "Choose logs to include in sink" panel, filter for your data source’s audit logs. Follow the below examples to create a filter for your data source configuration.
2. Create a Pub/Sub Subscription that receives messages from the Pub/Sub Topic. The Agentless Gateway service will pull the audit logs from this subscription.

The gcloud command line tool - linked to a project owner service account - may also be used to create the Log Router Sink and export the audit logs to a Pub/Sub Topic.

Routing logs of a single cluster to a Pub/Sub Topic (One-to-One)

To send audit logs of a single cluster to a single Pub/Sub topic, use the filter below while creating the Log Router Sink. Replace the "<GCP-project-id>" and <cluster_id> with your own values in the filters.

```
resource.labels.cluster_id="<cluster_id>" AND (logName="projects/<GCP-project-id>/logs/cloud
```

Routing logs of multiple clusters to a Pub/Sub Topic (Many-to-One)

```
(resource.labels.cluster_id="<cluster_id>"
OR
resource.labels.cluster_id="<cluster_id>")
AND
(logName="projects/<GCP-project-id>/logs/cloudaudit.googleapis.com%2Fdata_access" OR logName
```

> **Note:**
>
> Due to the volume and frequency, it is recommended to build an exclusion filter to exclude internal traffic coming from the "alloydbadmin" user. For example, the following addition to the sink will filter out any log by this user included in the textPayload section: AND NOT (textPayload:"user=alloydbadmin")

Creating a Pub/Sub Subscription

Create a Pub/Sub subscription by following the instructions at Create pull subscriptions. For more information on Pub/Sub Topics and Subscriptions, see Overview of the Pub/Sub service and Create a topic.

> **Note:**
>
> - It is recommended to have one Agentless Gateway pull from a given Pub/Sub subscriptions. Multiple Agentless Gateways pulling from the same Pub/Sub subscription may result in the ingestion of partial data.
> - A single Pub/Sub subscription can store the audit logs of multiple instances of the same Server Type. However, the audit logs of different Server Types cannot be sent to the same Pub/Sub subscription.
> - If several instances of the same Server Type write audit data to a single Pub/Sub subscription, connecting any one of the data source assets or the Pub/Sub subscription asset will pull all the audit data from the subscription. The Pub/Sub asset will stay connected to the gateway and will pull all the audit data until all the data source assets using it are disconnected.

Network Prerequisites

Below is an overview of the necessary requirements to enable the Agentless Gateway to communicate with data sources and other components across the network. Please ensure these are completed by a Network Administrator.

Agentless Gateway Firewall Settings

DSF reads audit events from Google's Pub/Sub service running on the Google cloud. Ensure the Agentless Gateway host is able to access the Pub/Sub subscription. Note that:

- The Agentless Gateway host must be able to reach the Google Cloud Pub/Sub API endpoint `pubsub.googleapis.com`. If the gateway is configured with dynamic subscribers (a non-default setting), it must also reach `monitoring.googleapis.com`.
- The firewall needs to allow traffic from the Agentless Gateway on several outbound ports to enable communication with the Pub/Sub subscription:
  - Ports 80 and 443: for HTTP and HTTPS requests to the Pub/Sub subscription.

Database Prerequisites

Below are the requirements for granting database permissions for audit log retrieval. Please ensure these are completed by a Database Administrator.

- To create the pgaudit extension and enable audit, you will need to connect to the primary instance as a user with the "alloydbsuperuser" role. Please obtain the credentials of a user with this role from the DBA.

✔STEP 2: Enabling Audit on the Data Source

Once the required permissions and information have been obtained, please complete the following steps to enable audit on the data source.

Enable Audit Logging

The Cloud AlloyDB auditing facility provides two distinct types of messages:

1. Admin Activity audit logs: Includes "admin write" operations that write metadata or configuration information. The Admin Activity audit logs are enabled by default and cannot be disabled.
2. Data Access audit logs: Includes "admin read" operations that read metadata or configuration information. It also includes "data read" and "data write" operations that read or write user provided data. To receive Data Access audit logs, you must explicitly enable them. Follow these steps to enable them:

1. a. Navigate to the "IAM and Admin" page and select "Audit Logs" tab on GCP console.
   b. Find the AlloyDB API Service in the list and enable “Admin Read”, “Data Read”, and “Data Write”. For more information, see AlloyDB for PostgreSQL audit logging.

Once the Data Access audit logs are enabled you can go ahead and modify the audit configuration for your specific cluster.

Enable Auditing on AlloyDB for PostgreSQL Cluster

Auditing can be configured on the AlloyDB for PostgreSQL Cluster by modifying the database flags for each instance. For more information, see Configure an instance's database flags. For example, to enable audit on the cluster, execute the following steps from the Google Cloud console:

1. In the Google Cloud AlloyDB console, navigate to the Clusters page.
2. Click on the desired cluster in the Resource Name column.

3. In the Overview page, go to Instances in your cluster, select an instance, and then click "Edit".

4. To add a database flag to your instance click "Add a Database flag".

5. Set the following flags:

Flag	Value	Notes
alloydb.enable_pgaudit	on	
log_connections	on	
log_disconnections	on	
log_error_verbosity	verbose	
log_hostname	on	
pgaudit.log	all	Setting the **pgaudit.log** parameter

		to **all** will log all events. To set up a more granular Audit Policy, see pgAudit documentation.

For DSF version 15.0 and newer set this additional flag:

Flag	Value	Notes
log_line_prefix	SONAR_AUDIT=1|TIMESTAMP=%m|APPLICATION_NAME=%a|USER=%u|DATABASE=%d|REMOTE_HOST_AND_PORT=%r|SQL_STATE=%e|SESSION_ID=%c|SESSION_START=%s|PROCESS_ID=[%p]|VIRTUAL_TRANSACTION_ID=%v|TRANSACTION_ID=%x|	It is required to use the log_line_prefix exactly as shown. Any modifications may cause system errors.

6. Click "Update instance" when finished. A text box will appear notifying you that your instance will need to be restarted, click "confirm and restart".
7. After enabling the database flags, log in to the primary instance with a user that is assigned the "alloydbsuperuser" role.
8. Confirm the pgaudit plugin has been installed by running the following command:

```
SHOW shared_preload_libraries;
```

The output should include "pgaudit" in the result (Note: there can be other values in the list of shared_preload_libraries depending on the cluster configuration)

```
Shared_preload_libraries
-------------------------
pgaudit
```

9. Create the extension by running the following command:

```
CREATE EXTENSION pgaudit;
```

10. Verify the extension was created successfully by running the following command:

```
postgres=> \dx
                          List of installed extensions
          Name           | Version |    Schema    |            Description
```

```
------------------------+---------+-----------+---------------------------------
 google_columnar_engine | 1.0     | public     |
 google_db_advisor      | 1.0     | public     |
 hypopg                 | 1.3.2   | public     |
 pgaudit                | 1.7     | public     | provides auditing functionality
 plpgsql                | 1.0     | pg_catalog | PL/pgSQL procedural language
(5 rows)
```

11. To verify audit configuration was set properly you can run the following command on your instance:

```
SELECT name, setting FROM pg_settings WHERE name in ('log_disconnections','log_connection
```

This should be the output:

```
        name          | setting
---------------------+---------
 log_connections      | on
 log_disconnections   | on
 log_error_verbosity | verbose
 log_hostname         | on
 log_line_prefix      | SONAR_AUDIT=1|TIMESTAMP=%m|APPLICATION_NAME=%a|USER=%u|DATABASE=%d
 pgaudit.log          | all
```

Enable Slow Query Auditing (Optional)

Slow query logs audit SQL statements that run longer than a specified duration. This duration is set using the flag "log_min_duration_statement" under the "**Configuration Options**" section in addition to the audit flags enabled in the previous step.

Flag	Value	Notes
log_min_duration_statement	<value>	<value> is your desired threshold in milliseconds and may be updated as desired

Note:

- **Turning on Slow Query Monitoring is supported from DSF version 15.0 and onwards only for AlloyDB clusters that have the log_line_prefix flag set.**
- Turning on Slow Query Monitoring will not interfere with standard audit collection.
- Queries that exceed the minimum duration threshold will be mapped to a separate collection called sonargd.slow_query.
- Make sure the **log_statement** parameter is set to **None**. Any modifications may cause slow query

logs to not be ingested.
- This step is **optional** and required only if you choose to audit slow query logs.
- The following setting should be done IN ADDITION to all previous steps.

✔STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

GCP AlloyDB PostgreSQL Cluster Asset Specifications

GCP AlloyDB PostgreSQL Asset Specifications

GCP Pub Sub Asset Specifications

GCP Asset Specifications

Note:

Audit logs of multiple AlloyDB clusters can be routed to one Pub/Sub topic. If you have this type of setup, complete these steps:

1. Create unique assets for each AlloyDB cluster and its corresponding instances. Only one asset is required for the Pub/Sub subscription asset.
2. Connecting the Pub/Sub asset will pull the audit data for all the clusters. The Pub/Sub asset will stay connected to the Agentless Gateway and pull all the audit data until all its associated cluster and instance assets are disconnected from the Agentless Gateway.

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.

2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.

3. In the Data Source Type dropdown menu, select a data source.

4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.

5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click Advanced.

6. Under Monitoring, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.

7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the Adding Assets via Unified Settings Console (USC) documentation.

Discover Assets using Cloud Accounts in USC

The Cloud Accounts tab allows you to run the Discovery playbook at any time to automatically discover and onboard new assets. Follow the instructions below to add a new cloud account and run Discovery via USC.

To add a new cloud account:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Cloud Accounts tab to view the cloud accounts of the DSF Hub. Click **Add** to open the Add Cloud Account form.
3. In the Cloud Account Type dropdown menu, select Amazon, Azure, or Google Cloud.
4. Specific configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. Under Connection, select the authentication mechanism that the Agentless Gateway will use to connect to the cloud account.
   - For Amazon: in the Auth Mechanism dropdown menu, select Profile, Key, IAM Role or Default.
   - For Azure: in the Auth Mechanism dropdown menu, select Client Secret, Auth File, or Managed Identity.
   - For Google Cloud: in the Auth Mechanism dropdown menu, select Default or Service Account.
6. Depending on the selected cloud account type and authentication mechanism, different fields will be displayed under the Connection section. Fill them out accordingly.
7. Under Monitoring, select the Agentless Gateway that will own the cloud account in the Gateway field, then click **Save**.

To run Discovery in USC:

1. Locate the desired cloud account asset in USC as described above.
2. Click on the kebab menu icon to the right of the cloud account on which you want to run the Discovery feature.
3. Click **Start Discovery**. Discovery will start running in the background.
4. After Discovery has run successfully, locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on importing data source assets using cloud accounts in USC, see the Discovering Data

Sources in DSF Hub Cloud Accounts using USC documentation.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

1. From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specification documentation as a guide for filling in the necessary field values.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Management** > **Audit Actions** > **Connect Gateway** to start collecting audit data.

For more information, please see the Adding Assets via the Import Assets Page documentation.

Note:

- Asset template spreadsheet name for AlloyDB for PostgreSQL: **GOOGLECLOUD_ALLOYDB_POSTGRESQL_template.xlsx**
- If you are sending audit logs of multiple AlloyDB clusters to a single Pub/Sub, filter for the assets and click on the "Management" button displayed above the assets → Audit Actions → Connect/Disconnect Gateway to connect or disconnect all assets simultaneously.

Discover Assets using Cloud Accounts in the Assets Dashboard

Note:

This section can be skipped if the data source asset(s) have already been onboarded using spreadsheets in the previous section.

After configuring and onboarding a cloud account asset with the necessary permissions, the Asset Discovery feature can find and onboard related data source assets with just a few clicks.

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download.**
3. Use the Asset Specifications documentation as a guide for completing the asset template spreadsheet for the

cloud account asset.

4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **Validate All**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the cloud account asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Discovery**. Choose the data source type to be discovered to start the Discovery playbook.
9. After the Discovery playbook has run successfully, locate the desired data source asset.
10. Click the kebab menu to the right of the asset, then click **Management** > **Audit Actions** > **Connect Gateway** to start collecting audit data.

For more information on discovering assets, please see the Adding Assets via Asset Discovery documentation.

> **Note:**
>
> Asset template name for Google Cloud Account Asset; choose any ONE of the templates listed below based on the type of authentication mechanism:
>
> - **GOOGLECLOUD_SERVICE_ACCOUNT_template.xlsx**
> - **GOOGLECLOUD_DEFAULT_template.xlsx**

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see Using DSF Open APIs.

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via https://supportportal.thalesgroup.com/csm.

On the Agentless Gateway host(s) please review the following:

DSF Hub versions 15.3 and newer :

- Gateway log file: $JSONAR_LOGDIR/gateway/syslog/alloydb_postgresql_pubsub.log
- Gateway pull log file: `$JSONAR_LOGDIR/gateway/pull/google-pubsub/gateway-pull.log`

- Run the following command to verify that the required services are active:

```
systemctl status -l gateway-pull@google-pubsub.service
systemctl status -l sonarrsyslog
```

DSF Hub versions 4.11 to 15.2 :

- Gateway log file: $JSONAR_LOGDIR/gateway/syslog/alloydb_postgresql_pubsub.log
- SonarGD log file: $JSONAR_LOGDIR/sonargd/sonargd.log

- Run the following command to verify the status of the services:

```
systemctl status -l sonargd
systemctl status -l sonarrsyslog
```

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via https://supportportal.thalesgroup.com/csm. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

- Using DSF Open APIs: A guide to onboarding and managing assets via DSF Open APIs.
- Adding assets via Unified Settings Console (USC): Instructions to add and edit data source assets via the USC.
- Adding Assets via the Import Assets Page: Instructions to add data source assets via the Asset Dashboard.
- Using the Data Security Fabric Portal: An overview of the Data Security Fabric (DSF) portal.
- Data Security Coverage Tool: Systems and version compatibility with DSF Hub.

Document generated by Confluence on Jun 19, 2026 01:21

Atlassian

BigQuery Onboarding Steps

This topic reviews the steps required to audit activity on GCP BigQuery databases and send that information to DSF Hub for auditing, analysis, detecting and preventing security events.

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The following main topics are included:

- An overview of the available scope for the data source, given its native audit data.

- A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.

- Instructions on how to enable audit on the data source and collect data using DSF Hub.

- Initial troubleshooting steps and technical support information.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps. Please click on each of the steps below to display the content.

STEP 1: Onboarding Prerequisites

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

Permissions for Google Cloud Account Principal

Grant the following IAM roles to a cloud account principal:

- **Logging Admin** and **Project IAM Admin** roles: for enabling Data Access logs, viewing logs and creating a Log Router Sink.
- **Pub/Sub Admin** role: for creating a Pub/Sub Topic and Subscription.
- **Service Account Admin** and optionally **Service Account Key Admin** roles: for creating a service account and generating keys (if using "service_account" auth mechanism). See Permissions and roles for more information on the list of individual permissions contained in each predefined role.

Creating a Google Cloud Service Account

DSF supports the following two authentication mechanisms for accessing Google Cloud resources. Note that both authentication mechanisms make use of a Service Account. The process for creating a Service Account is described in Creating and managing service accounts.

- **Service Account:** to use the `service_account` authentication mechanism, a JSON-formatted key file needs to be generated holding the credentials that the Agentless Gateway will use to access audit logs from the Pub/Sub subscription. Ensure that the key file is copied to the Agentless Gateway host and it is owned by the `sonarw` user. For more information on how to create a service account key, see Creating and managing service account keys.

- **Default:** to use the `default` authentication mechanism, attach the Service Account created above to the Agentless Gateway hosted on a Google Cloud Compute Engine virtual machine instance. For more information on how to attach a Service Account to a GCP Compute Engine virtual machine, see Change the attached service account.

The Service Account should be granted the following roles:

- Pub/Sub Subscriber
- Pub/Sub Viewer
- Viewer

> **Note:**
>
> It is recommended to keep both the Service Account and the Agentless Gateway Google Cloud VM instance within the same Google Cloud Project.

Google Cloud Log Routing Overview: Log Router Sinks, Pub/Sub Topics and Subscriptions

Log Router Sinks control how Logging routes logs in Google Cloud. Using sinks, you can route audit logs to one or more supported destinations. Please follow the instructions in Route logs to supported destinations. The following summarizes the steps to be completed:

1. Create a Log Router sink that sends the audit logs to a new or existing Pub/Sub Topic as the destination service.
   a. In the "Choose logs to include in sink" panel, filter for your data source's audit logs. Follow the below examples to create a filter for your data source configuration.
2. Create a Pub/Sub Subscription that receives messages from the Pub/Sub Topic. The Agentless Gateway service will pull the audit logs from this subscription.

The gcloud command line tool - linked to a project owner service account - may also be used to create the Log Router Sink and export the audit logs to a Pub/Sub Topic.

To send BigQuery audit logs to a Pub/Sub Topic, use the filter below while creating the Log Router Sink.

```
resource.type="bigquery_resource"
```

Creating a Pub/Sub Subscription

Create a Pub/Sub subscription by following the instructions at Create pull subscriptions. For more information on Pub/Sub Topics and Subscriptions, see Overview of the Pub/Sub service and Create a topic.

> **Note:**
>
> - It is recommended to have one Agentless Gateway pull from a given Pub/Sub subscriptions. Multiple Agentless Gateways pulling from the same Pub/Sub subscription may result in the ingestion of partial data.
> - A single Pub/Sub subscription can store the audit logs of multiple instances of the same Server Type.

However, the audit logs of different Server Types cannot be sent to the same Pub/Sub subscription.

- If several instances of the same Server Type write audit data to a single Pub/Sub subscription, connecting any one of the data source assets or the Pub/Sub subscription asset will pull all the audit data from the subscription. The Pub/Sub asset will stay connected to the gateway and will pull all the audit data until all the data source assets using it are disconnected.

Network Prerequisites

Below is an overview of the necessary requirements to enable the Agentless Gateway to communicate with data sources and other components across the network. Please ensure these are completed by a Network Administrator.

Agentless Gateway Firewall Settings

DSF reads audit events from Google's Pub/Sub service running on the Google cloud. Ensure the Agentless Gateway host is able to access the Pub/Sub subscription. Note that:

- The Agentless Gateway host must be able to reach the Google Cloud Pub/Sub API endpoint `pubsub.googleapis.com`. If the gateway is configured with dynamic subscribers (a non-default setting), it must also reach `monitoring.googleapis.com`.
- The firewall needs to allow traffic from the Agentless Gateway on several outbound ports to enable communication with the Pub/Sub subscription:
  - Ports 80 and 443: for HTTP and HTTPS requests to the Pub/Sub subscription.

✔STEP 2: Enabling Audit on the Data Source

Data Access audit logs for BigQuery are enabled by default. No additional action is required, please proceed to the next step. For more information, please see Enable Data Access audit logs.

✔STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

GCP BigQuery Asset Specifications

GCP Pub Sub Asset Specifications

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.
3. In the Data Source Type dropdown menu, select a data source.
4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click Advanced.
6. Under Monitoring, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.
7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the Adding Assets via Unified Settings Console (USC) documentation.

> **Note:**
>
> To monitor slow queries, enter an integer value in the **Duration Threshold** field to specify the execution time in milliseconds after which a query is flagged as slow.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

1. From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specification documentation as a guide for filling in the necessary field values.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.

6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Management** > **Audit Actions** > **Connect Gateway** to start collecting audit data.

For more information, please see the Adding Assets via the Import Assets Page documentation.

> **Note:**
>
> Asset template name for BigQuery: ***GOOGLECLOUD_BIGQUERY_template.xlsx***
>
> - To monitor slow queries, add the **duration_threshold** column to the BIG QUERY asset row and set an integer value for the query execution time in milliseconds. Queries that exceed this threshold will be flagged as slow.

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see Using DSF Open APIs.

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via https://supportportal.thalesgroup.com/csm.

On the Agentless Gateway host(s) please review the following:

DSF Hub versions 15.3 and newer :

- Gateway log file: ${JSONAR_LOGDIR}/gateway/syslog/bigquery_pubsub.log
- Gateway pull log file: `$JSONAR_LOGDIR/gateway/pull/google-pubsub/gateway-pull.log`

- Run the following command to verify that the required services are active:

```
systemctl status -l gateway-pull@google-pubsub.service
systemctl status -l sonarrsyslog
```

DSF Hub versions 4.11 to 15.2:

- Gateway log file: ${JSONAR_LOGDIR}/gateway/syslog/bigquery_pubsub.log
- sonargd log file : ${JSONAR_LOGDIR}/sonargd/sonargd.log

- Run the following command to verify the status of the services:

```
systemctl status -l sonargd
systemctl status -l sonarrsyslog
```

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via https://supportportal.thalesgroup.com/csm. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

- Using DSF Open APIs: A guide to onboarding and managing assets via DSF Open APIs.
- Adding assets via Unified Settings Console (USC): Instructions to add and edit data source assets via the USC.
- Adding Assets via the Import Assets Page: Instructions to add data source assets via the Asset Dashboard.
- Using the Data Security Fabric Portal: An overview of the Data Security Fabric (DSF) portal.
- Data Security Coverage Tool: Systems and version compatibility with DSF Hub.

Document generated by Confluence on Jun 19, 2026 01:21

Atlassian

Bigtable Onboarding Steps

This topic reviews the steps required to audit activity on Cloud BigTable databases and send that information to DSF Hub for auditing, analysis, detecting and preventing security events.

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The following main topics are included:

- An overview of the available scope for the data source, given its native audit data.
- A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.
- Instructions on how to enable audit on the data source and collect data using DSF Hub.
- Initial troubleshooting steps and technical support information.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps. Please click on each of the steps below to display the content.

STEP 1: Onboarding Prerequisites

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

Permissions for Google Cloud Account Principal

Grant the following IAM roles to a cloud account principal:

- **Logging Admin** and **Project IAM Admin** roles: for enabling Data Access logs, viewing logs and creating a Log Router Sink.
- **Pub/Sub Admin** role: for creating a Pub/Sub Topic and Subscription.
- **Service Account Admin** and optionally **Service Account Key Admin** roles: for creating a service account and generating keys (if using "service_account" auth mechanism). See Permissions and roles for more information on the list of individual permissions contained in each predefined role.

Creating a Google Cloud Service Account

DSF supports the following two authentication mechanisms for accessing Google Cloud resources. Note that both authentication mechanisms make use of a Service Account. The process for creating a Service Account is described in Creating and managing service accounts.

- **Service Account:** to use the `service_account` authentication mechanism, a JSON-formatted key file needs to be generated holding the credentials that the Agentless Gateway will use to access audit logs from the Pub/Sub subscription. Ensure that the key file is copied to the Agentless Gateway host and it is owned by the `sonarw` user. For more information on how to create a service account key, see Creating and managing service account keys.

- **Default:** to use the `default` authentication mechanism, attach the Service Account created above to the Agentless Gateway hosted on a Google Cloud Compute Engine virtual machine instance. For more information on how to attach a Service Account to a GCP Compute Engine virtual machine, see Change the attached service account.

The Service Account should be granted the following roles:

- Pub/Sub Subscriber

- Pub/Sub Viewer

- Viewer

Note:

> It is recommended to keep both the Service Account and the Agentless Gateway Google Cloud VM instance within the same Google Cloud Project.

Google Cloud Log Routing Overview: Log Router Sinks, Pub/Sub Topics and Subscriptions

Log Router Sinks control how Logging routes logs in Google Cloud. Using sinks, you can route audit logs to one or more supported destinations. Please follow the instructions in Route logs to supported destinations. The following summarizes the steps to be completed:

1. Create a Log Router sink that sends the audit logs to a new or existing Pub/Sub Topic as the destination service.
    a. In the "Choose logs to include in sink" panel, filter for your data source’s audit logs. Follow the below examples to create a filter for your data source configuration.
2. Create a Pub/Sub Subscription that receives messages from the Pub/Sub Topic. The Agentless Gateway service will pull the audit logs from this subscription.

The gcloud command line tool - linked to a project owner service account - may also be used to create the Log Router Sink and export the audit logs to a Pub/Sub Topic.

Use the filter below while creating the Log Router Sink to send Bigtable audit logs to a Pub/Sub Topic, replacing <GCP-project-id> with your own value.

```
resource.type="audited_resource" resource.labels.service=("bigtable.googleapis.com" OR "bigt
logName=("projects/<GCP-project-id>/logs/cloudaudit.googleapis.com%2Factivity" OR "projects/
```

Creating a Pub/Sub Subscription

Create a Pub/Sub subscription by following the instructions at Create pull subscriptions. For more information on Pub/Sub Topics and Subscriptions, see Overview of the Pub/Sub service and Create a topic.

> **Note:**
>
> - It is recommended to have one Agentless Gateway pull from a given Pub/Sub subscriptions. Multiple Agentless Gateways pulling from the same Pub/Sub subscription may result in the ingestion of partial data.
> - A single Pub/Sub subscription can store the audit logs of multiple instances of the same Server Type. However, the audit logs of different Server Types cannot be sent to the same Pub/Sub subscription.
> - If several instances of the same Server Type write audit data to a single Pub/Sub subscription, connecting any one of the data source assets or the Pub/Sub subscription asset will pull all the audit

> data from the subscription. The Pub/Sub asset will stay connected to the gateway and will pull all the audit data until all the data source assets using it are disconnected.

Network Prerequisites

Below is an overview of the necessary requirements to enable the Agentless Gateway to communicate with data sources and other components across the network. Please ensure these are completed by a Network Administrator.

Agentless Gateway Firewall Settings

DSF reads audit events from Google's Pub/Sub service running on the Google cloud. Ensure the Agentless Gateway host is able to access the Pub/Sub subscription. Note that:

- The Agentless Gateway host must be able to reach the Google Cloud Pub/Sub API endpoint `pubsub.googleapis.com`. If the gateway is configured with dynamic subscribers (a non-default setting), it must also reach `monitoring.googleapis.com`.

- The firewall needs to allow traffic from the Agentless Gateway on several outbound ports to enable communication with the Pub/Sub subscription:

  - Ports 80 and 443: for HTTP and HTTPS requests to the Pub/Sub subscription.

✔STEP 2: Enabling Audit on the Data Source

Once the required permissions and information have been obtained, please complete the following steps to enable audit on the data source.

Enable Audit Logging

The Cloud Bigtable auditing facility provides two distinct types of messages:

- Admin Activity audit logs: includes "admin write" operations that write metadata or configuration information. These messages are enabled by default.
- Data Access audit logs: includes "admin read" operations that read metadata or configuration information, as well as "data read" and "data write" operations that read or write user provided data. These messages are disabled by default.

To enable Data Access audit logs, please complete the following:

1. From the Google Cloud console, navigate to IAM & Admin > Audit Logs.
2. In the Data Access audit logs configuration table, select Cloud Bigtable APIs service in the Service column.
3. In the Log Types tab that appears on the right, enable "Admin Read", "Data Read", and "Data Write". For more information, see the GCP Audit logs documentation.
4. Click Save.

For more information, please refer to the Bigtable audit logs overview.

✔STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

GCP Bigtable Asset Specifications

GCP Pub Sub Asset Specifications

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.

2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.

3. In the Data Source Type dropdown menu, select a data source.

4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.

5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click Advanced.

6. Under Monitoring, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.

7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the Adding Assets via Unified Settings Console (USC) documentation.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

1. From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specification documentation as a guide for filling in the necessary field values.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Management** > **Audit Actions** > **Connect Gateway** to start collecting audit data.

For more information, please see the Adding Assets via the Import Assets Page documentation.

> **Note:**
>
> Asset template name for Bigtable: **GOOGLECLOUD_BIGTABLE_template.xlsx**

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see Using DSF Open APIs.

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via https://supportportal.thalesgroup.com/csm.

On the Agentless Gateway host(s) please review the following:

DSF Hub Versions 15.3 and newer :

- Gateway log file: ${JSONAR_LOGDIR}/gateway/syslog/bigtable_pubsub.log
- Gateway pull log file: $JSONAR_LOGDIR/gateway/pull/google-pubsub/gateway-pull.log

- Run the following command to verify that the required services are active:

```
systemctl status -l gateway-pull@google-pubsub.service
systemctl status -l sonarrsyslog
```

DSF Hub Versions 4.11 to 15.2:

- Gateway log file: ${JSONAR_LOGDIR}/gateway/syslog/bigtable_pubsub.log

- sonargd log file : ${JSONAR_LOGDIR}/sonargd/sonargd.log

- Run the following command to verify the status of the services:

```
systemctl status -l sonargd
systemctl status -l sonarrsyslog
```

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via https://supportportal.thalesgroup.com/csm. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

- Using DSF Open APIs: A guide to onboarding and managing assets via DSF Open APIs.
- Adding assets via Unified Settings Console (USC): Instructions to add and edit data source assets via the USC.
- Adding Assets via the Import Assets Page: Instructions to add data source assets via the Asset Dashboard.
- Using the Data Security Fabric Portal: An overview of the Data Security Fabric (DSF) portal.
- Data Security Coverage Tool: Systems and version compatibility with DSF Hub.

Document generated by Confluence on Jun 19, 2026 01:21

Atlassian

Cloud SQL for MySQL Onboarding Steps

This topic reviews the steps required to audit activity on Cloud SQL for MySQL databases and send that information to DSF Hub for auditing, analysis, detecting and preventing security events.

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The following main topics are included:

- An overview of the available scope for the data source, given its native audit data.

- A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.

- Instructions on how to enable audit on the data source and collect data using DSF Hub.

- Initial troubleshooting steps and technical support information.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps. Please click on each of the steps below to display the content.

✔STEP 1: Onboarding Prerequisites

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

Permissions for Google Cloud Account Principal

Grant the following IAM roles to a cloud account principal:

- **Logging Admin** and **Project IAM Admin** roles: for enabling Data Access logs, viewing logs and creating a Log Router Sink.
- **Pub/Sub Admin** role: for creating a Pub/Sub Topic and Subscription.
- **CloudSQL Admin** role: for modifying database-specific flags.
- **Service Account Admin** and optionally **Service Account Key Admin** roles: for creating a service account and generating keys (if using "service_account" auth mechanism). See Permissions and roles for more information on the list of individual permissions contained in each predefined role.

Creating a Google Cloud Service Account

DSF supports the following two authentication mechanisms for accessing Google Cloud resources. Note that both authentication mechanisms make use of a Service Account. The process for creating a Service Account is described in Creating and managing service accounts.

- **Service Account:** to use the `service_account` authentication mechanism, a JSON-formatted key file needs to be generated holding the credentials that the Agentless Gateway will use to access audit logs from the Pub/Sub subscription. Ensure that the key file is copied to the Agentless Gateway host and it is owned by the `sonarw` user. For more information on how to create a service account key, see Creating and managing service account keys.
- **Default:** to use the `default` authentication mechanism, attach the Service Account created above to the Agentless Gateway hosted on a Google Cloud Compute Engine virtual machine instance. For more information on how to attach a Service Account to a GCP Compute Engine virtual machine, see Change the attached service account.

The Service Account should be granted the following roles:

- Pub/Sub Subscriber

- Pub/Sub Viewer
- Viewer

> **Note:**
>
> It is recommended to keep both the Service Account and the Agentless Gateway Google Cloud VM instance within the same Google Cloud Project.

Google Cloud Log Routing Overview: Log Router Sinks, Pub/Sub Topics and Subscriptions

Log Router Sinks control how Logging routes logs in Google Cloud. Using sinks, you can route audit logs to one or more supported destinations. Please follow the instructions in Route logs to supported destinations. The following summarizes the steps to be completed:

1. Create a Log Router sink that sends the audit logs to a new or existing Pub/Sub Topic as the destination service.
   a. In the "Choose logs to include in sink" panel, filter for your data source’s audit logs. Follow the below examples to create a filter for your data source configuration.
2. Create a Pub/Sub Subscription that receives messages from the Pub/Sub Topic. The Agentless Gateway service will pull the audit logs from this subscription.

The gcloud command line tool - linked to a project owner service account - may also be used to create the Log Router Sink and export the audit logs to a Pub/Sub Topic.

> **Note:**
>
> - Cloud SQL for MySQL supports standard audit monitoring and slow query audit monitoring. To enable auditing of both types simultaneously on the same instance, configure a separate Log Router Sink, Pub/Sub Topic and Pub/Sub Subscription for each audit type.
> - Due to its volume and frequency, it is recommended to build an exclusion filter that filters out internal traffic coming from the "__google_connectivity_prober" user.
> - If you choose to discover your database assets, it is recommended to:
>   1. Start the Log Router Sink name with "sonar". For example, "sonar-gcp-mysql-sink".
>   2. Associate the database_id with only one Log Router Sink.

Routing logs of a single instance to a Pub/Sub Topic (One-to-One)

To send the audit logs of a single instance to a single Pub/Sub topic, use the following inclusion filter while creating the Log Router Sink. Replace the <GCP-project-id> and <mysql-instance-name> with your own values.

```
resource.type="cloudsql_database"
resource.labels.database_id="<GCP-project-id>:<mysql-instance-name>"
logName="projects/<GCP-project-id>/logs/cloudsql.googleapis.com%2Fmysql-general.log"
```

Routing logs of multiple instances to a Pub/Sub Topic (Many-to-One)

To send the audit logs of multiple instances to a single Pub/Sub topic, use an OR operator in the inclusion filter while creating the Log Router Sink. Replace the <GCP-project-id> and instance names with your own values.

```
resource.type="cloudsql_database"
resource.labels.database_id="<GCP-project-id>:<mysql-INSTANCE1-name>" OR resource.labels.dat
logName="projects/<GCP-project-id>/logs/cloudsql.googleapis.com%2Fmysql-general.log"
```

Routing Slow Query logs of a single instance to a Pub/Sub Topic (Optional)

To send the slow query audit logs of a single instance to a single Pub/Sub topic, use the following inclusion filter while creating the Log Router Sink. Replace the <GCP-project-id> and <mysql-instance-name> with your own values.

```
resource.type="cloudsql_database"
resource.labels.database_id="<GCP-project-id>:<mysql-instance-name>"
logName="projects/<GCP-project-id>/logs/cloudsql.googleapis.com%2Fmysql-slow.log"
```

Creating a Pub/Sub Subscription

Create a Pub/Sub subscription by following the instructions at Create pull subscriptions. For more information on Pub/Sub Topics and Subscriptions, see Overview of the Pub/Sub service and Create a topic.

Note:

- It is recommended to have one Agentless Gateway pull from a given Pub/Sub subscriptions. Multiple Agentless Gateways pulling from the same Pub/Sub subscription may result in the ingestion of partial data.
- A single Pub/Sub subscription can store the audit logs of multiple instances of the same Server Type. However, the audit logs of different Server Types cannot be sent to the same Pub/Sub subscription.
- If several instances of the same Server Type write audit data to a single Pub/Sub subscription, connecting any one of the data source assets or the Pub/Sub subscription asset will pull all the audit data from the subscription. The Pub/Sub asset will stay connected to the gateway and will pull all the audit data until all the data source assets using it are disconnected.

Network Prerequisites

Below is an overview of the necessary requirements to enable the Agentless Gateway to communicate with data sources and other components across the network. Please ensure these are completed by a Network

Administrator.

Agentless Gateway Firewall Settings

DSF reads audit events from Google's Pub/Sub service running on the Google cloud. Ensure the Agentless Gateway host is able to access the Pub/Sub subscription. Note that:

- The Agentless Gateway host must be able to reach the Google Cloud Pub/Sub API endpoint `pubsub.googleapis.com`. If the gateway is configured with dynamic subscribers (a non-default setting), it must also reach `monitoring.googleapis.com`.

- The firewall needs to allow traffic from the Agentless Gateway on several outbound ports to enable communication with the Pub/Sub subscription:

  - Ports 80 and 443: for HTTP and HTTPS requests to the Pub/Sub subscription.

✔STEP 2: Enabling Audit on the Data Source

Once the required permissions and information have been obtained, please complete the following steps to enable audit on the data source.

Enable Audit Logging

Enable Auditing on Cloud SQL

The Cloud SQL service provides three distinct types of messages:

- System Event audit logs: these messages identify automated Google Cloud actions that modify the configuration of resources.
- Admin Activity audit logs: these messages include "admin write" operations that write metadata or configuration information.
- Data Access audit logs: these messages include "admin read" operations that read metadata or configuration information. They also include "data read" and "data write" operations that read or write user provided data.

System Event and Admin Activity audit logs are enabled by default. Data Access audit logs are disabled by default. To enable them, complete the following steps:

1. From the Google Cloud console, navigate to IAM & Admin > Audit Logs.
2. In the Data Access audit logs configuration table, select Cloud SQL service in the Service column.
3. In the Log Types tab that appears on the right, enable “Admin Read”, “Data Read”, and “Data Write”. For more information, see Available audit logs.
4. Click Save.

Enable Auditing on Cloud SQL for MySQL instance

Auditing can be configured on the Cloud SQL for MySQL instance by modifying the database flags. For more information, refer to the Google Cloud documentation: Configuring database flags. To enable audit on the instance perform the following steps from the Google Cloud console:

1. Navigate to the left sidebar in Google Home console.
2. Click on “SQL”.
3. Click on your Cloud SQL for MySQL instance.
4. Click “Edit”.
5. Under the “Configuration options” section, click “Flags”.
6. Set the following flags and click Save when finished:

    - log_output: FILE

    - general_log: On

Enable Auditing of Slow Query Logs (Optional)

Slow query logs audit SQL statements that run longer than a specified duration. To enable Slow Query logging for your instance set these additional flags under the "Configuration options” section in addition to the audit flags enabled in the previous step:

- slow_query_log: On
- long_query_time: x.xx (a user-defined threshold)

✔STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

GCP MySQL Asset Specifications

GCP Pub Sub Asset Specifications

GCP Asset Specifications

> **Note:**
>
> **Many-to-one Setup**
>
> Audit logs of multiple Cloud SQL for MySQL instances may be ingested with one Pub/Sub topic by completing the following steps:
>
> 1. Create separate assets for each MySQL instance. Only one Pub/Sub log aggregator asset is required.
> 2. Connect the Pub/Sub asset to the Agentless Gateway to pull the audit data for all the MySQL instances.

The Pub/Sub asset will remain connected to the gateway and pull all of the audit data being sent to it until all of the instance assets are disconnected.

Standard and Slow Query Monitoring

Cloud SQL for MySQL supports both standard audit monitoring and slow query audit monitoring. To enable auditing of both types simultaneously on the same instance, complete the following steps:

1. Create separate log aggregator assets for each Pub/Sub audit type with the appropriate Audit Type value. See more details under your desired onboarding method.
   a. Because only one of the Pub/Sub assets may be defined in the "Logs Destination Asset ID" (logs_destination_asset_id) field of the Cloud SQL for MySQL asset, the other "standalone" Pub/Sub asset must have its "Content Type" (content_type) field set to "GCP MYSQL".
2. Connect each Pub/Sub to the Agentless Gateway to ingest both standard and slow query audit logs.

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.

2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.

3. In the Data Source Type dropdown menu, select a data source.

4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.

5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click Advanced.

6. Under Monitoring, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.

7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the Adding Assets via Unified Settings Console (USC) documentation.

Note:

While creating the Pub/Sub asset(s), select the appropriate Audit Type for your configuration:

- For standard audit monitoring: "MySQL"
- For slow query audit monitoring: "GCP MySQL Slow"

Discover Assets using Cloud Accounts in USC

The Cloud Accounts tab allows you to run the Discovery playbook at any time to automatically discover and onboard new assets. Follow the instructions below to add a new cloud account and run Discovery via USC.

To add a new cloud account:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Cloud Accounts tab to view the cloud accounts of the DSF Hub. Click **Add** to open the Add Cloud Account form.
3. In the Cloud Account Type dropdown menu, select Amazon, Azure, or Google Cloud.
4. Specific configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. Under Connection, select the authentication mechanism that the Agentless Gateway will use to connect to the cloud account.
   - For Amazon: in the Auth Mechanism dropdown menu, select Profile, Key, IAM Role or Default.
   - For Azure: in the Auth Mechanism dropdown menu, select Client Secret, Auth File, or Managed Identity.
   - For Google Cloud: in the Auth Mechanism dropdown menu, select Default or Service Account.
6. Depending on the selected cloud account type and authentication mechanism, different fields will be displayed under the Connection section. Fill them out accordingly.
7. Under Monitoring, select the Agentless Gateway that will own the cloud account in the Gateway field, then click **Save**.

To run Discovery in USC:

1. Locate the desired cloud account asset in USC as described above.
2. Click on the kebab menu icon to the right of the cloud account on which you want to run the Discovery feature.
3. Click **Start Discovery**. Discovery will start running in the background.
4. After Discovery has run successfully, locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on importing data source assets using cloud accounts in USC, see the Discovering Data Sources in DSF Hub Cloud Accounts using USC documentation.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

1. From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.

2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specification documentation as a guide for filling in the necessary field values.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Management** > **Audit Actions** > **Connect Gateway** to start collecting audit data.

For more information, please see the Adding Assets via the Import Assets Page documentation.

> **Note:**
>
> Asset template name for Cloud SQL for MySQL:
>
> - Standard audit monitoring: ***GOOGLECLOUD_CLOUDSQL_MYSQL_template.xlsx***
> - Slow query audit monitoring: ***GOOGLECLOUD_CLOUDSQL_MYSQL_SLOW_QUERY_template.xlsx***
>
> If you have routed the audit logs from multiple CloudSQL MySQL instances to a single Pub/Sub, filter for the data source assets in the search bar and click on the "Management" button icon displayed above the assets on the top right corner → Click on Audit Actions → Connect/Disconnect Gateway to connect or disconnect all assets at the same time.
>
> When running the Connect Gateway playbook on the Pub/Sub assets, you can select one of the possible **audit_type** options. Please select the most suitable audit_type for your configuration:
>
> - For standard audit monitoring: "MYSQL"
> - For slow query audit monitoring: "GCP_MYSQL_SLOW"

Discover Assets using Cloud Accounts in the Assets Dashboard

> **Note:**
>
> This section can be skipped if the data source asset(s) have already been onboarded using spreadsheets in the previous section.

After configuring and onboarding a cloud account asset with the necessary permissions, the Asset Discovery feature can find and onboard related data source assets with just a few clicks.

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download.**
3. Use the Asset Specifications documentation as a guide for completing the asset template spreadsheet for the

cloud account asset.

4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **Validate All**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the cloud account asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Discovery**. Choose the data source type to be discovered to start the Discovery playbook.
9. After the Discovery playbook has run successfully, locate the desired data source asset.
10. Click the kebab menu to the right of the asset, then click **Management** > **Audit Actions** > **Connect Gateway** to start collecting audit data.

For more information on discovering assets, please see the Adding Assets via Asset Discovery documentation.

> **Note:**
>
> Asset template name for Google Cloud Account Asset; choose any one of the templates listed below based on the type of authentication mechanism:
>
> - **GOOGLECLOUD_SERVICE_ACCOUNT_template.xlsx**
> - **GOOGLECLOUD_DEFAULT_template.xlsx**

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see Using DSF Open APIs.

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via https://supportportal.thalesgroup.com/csm.

On the Agentless Gateway host(s), please review the following:

DSF Hub versions 15.3 and newer :

- Gateway log file:
  - Standard audit monitoring: $JSONAR_LOGDIR/gateway/syslog/mysql_pubsub.log
  - Slow query audit monitoring: $JSONAR_LOGDIR/gateway/syslog/mysql_slow_pubsub.log
- Gateway pull log file: $JSONAR_LOGDIR/gateway/pull/google-pubsub/gateway-pull.log
- Run the following command to verify that the required services are active:

```
systemctl status -l gateway-pull@google-pubsub.service
systemctl status -l sonarrsyslog
```

DSF Hub versions 4.11 to 15.2:

- Gateway log file:
  - Standard audit monitoring: $JSONAR_LOGDIR/gateway/syslog/mysql_pubsub.log
  - Slow query audit monitoring: $JSONAR_LOGDIR/gateway/syslog/mysql_slow_pubsub.log
- SonarGD log file: $JSONAR_LOGDIR/sonargd/sonargd.log
- Run the following command to verify the status of the services:

```
systemctl status -l sonargd
systemctl status -l sonarrsyslog
```

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via https://supportportal.thalesgroup.com/csm. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

- Using DSF Open APIs: A guide to onboarding and managing assets via DSF Open APIs.
- Adding assets via Unified Settings Console (USC): Instructions to add and edit data source assets via the USC.
- Adding Assets via the Import Assets Page: Instructions to add data source assets via the Asset Dashboard.
- Using the Data Security Fabric Portal: An overview of the Data Security Fabric (DSF) portal.
- Data Security Coverage Tool: Systems and version compatibility with DSF Hub.

Cloud SQL for PostgreSQL Onboarding Steps

This topic reviews the steps required to audit activity on GCP PostgreSQL databases and send that information to DSF Hub for auditing, analysis, detecting and preventing security events.

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The following main topics are included:

- An overview of the available scope for the data source, given its native audit data.

- A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.
- Instructions on how to enable audit on the data source and collect data using DSF Hub.
- Initial troubleshooting steps and technical support information.

Audit Data Scope

The scope of audit data collection varies between data sources. Prior to enabling auditing, read through the auditing scope for this data source to identify any potential benefits or constraints. For more information on data collections, see Main Collections in sonargd Database.

- This source collects audit events in the **instance**, **exception**, and **session** collections. Optionally, events can also be collected in the **slow_query** collection.
- **Failed actions (exceptions):** When a failed action generates multiple log lines (`STATEMENT`, `LOCATION`, `ERROR`), only the `STATEMENT` line is processed to support multi-processing. As a result, the `Exception Description` is not captured. The `Error Code` is mapped to the `Exception Description` field instead.
- **Logon (session):** Logon events use metrics to aggregate multiple log lines into a single document.
- `Client IP`**,** `Client Hostname`**,** `Client Port`, and `Source Address` are only available in **session** and **exception** raw audit logs. For **instance** documents, these fields are mapped by correlating with the session's logon/logoff log via session ID, username, and database name, but only if the logon/logoff log arrives first. Otherwise, they default to `N/A`.
- Internal system operations, such as automatic table analysis and vacuuming, are excluded from audit logging. These logs are generated by an internal administrative user, do not include a statement parameter, and therefore do not match the metric filter. They cannot be initiated by external users.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps. Please click on each of the steps below to display the content.

STEP 1: Onboarding Prerequisites

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

Permissions for Google Cloud Account Principal

Grant the following IAM roles to a cloud account principal:

- **Logging Admin** and **Project IAM Admin** roles: for enabling Data Access logs, viewing logs and creating a Log Router Sink.
- **Pub/Sub Admin** role: for creating a Topic and Pub/Sub Subscription.
- **CloudSQL Admin** role: for modifying database-specific flags.
- **Service Account Admin** and optionally **Service Account Key Admin** roles: for creating a service account and generating keys (if using the "service_account" authentication mechanism). See Permissions and roles for more information on the list of individual permissions contained in each predefined role.

Creating a Google Cloud Service Account

DSF supports the following two authentication mechanisms for accessing Google Cloud resources. Note that both authentication mechanisms make use of a Service Account. The process for creating a Service Account is described in Creating and managing service accounts.

- **Service Account:** to use the `service_account` authentication mechanism, a JSON-formatted key file needs to be generated holding the credentials that the Agentless Gateway will use to access audit logs from the Pub/Sub subscription. Ensure that the key file is copied to the Agentless Gateway host and it is owned by the `sonarw` user. For more information on how to create a service account key, see Creating and managing service account keys.

- **Default:** to use the `default` authentication mechanism, attach the Service Account created above to the Agentless Gateway hosted on a Google Cloud Compute Engine virtual machine instance. For more information on how to attach a Service Account to a GCP Compute Engine virtual machine, see Change the attached service account.

The Service Account should be granted the following roles:

- Pub/Sub Subscriber

- Pub/Sub Viewer

- Viewer

> **Note:**
>
> It is recommended to keep both the Service Account and the Agentless Gateway Google Cloud VM instance within the same Google Cloud Project.

Google Cloud Log Routing Overview: Log Router Sinks, Pub/Sub Topics and Subscriptions

Log Router Sinks control how Logging routes logs in Google Cloud. Using sinks, you can route audit logs to one or more supported destinations. Please follow the instructions in Route logs to supported destinations. The following summarizes the steps to be completed:

1. Create a Log Router sink that sends the audit logs to a new or existing Pub/Sub Topic as the destination service.
    a. In the "Choose logs to include in sink" panel, filter for your data source’s audit logs. Follow the below examples to create a filter for your data source configuration.
2. Create a Pub/Sub Subscription that receives messages from the Pub/Sub Topic. The Agentless Gateway service will pull the audit logs from this subscription.

The gcloud command line tool - linked to a project owner service account - may also be used to create the Log Router Sink and export the audit logs to a Pub/Sub Topic.

> **Note:**
>
> - Due to its volume and frequency, it is recommended to build an exclusion filter to filter out the internal traffic coming from the "cloudsqladmin" user. For example: `AND NOT textPayload:"user=cloudsqladmin"`
>   This user is created and used by GCP for administrative purposes. For more details on actions generated by this user, see What is the cloudsqladmin database user?.
> - If you choose to discover your database assets, we recommend the following:
>   1. Start the sink name with "sonar" for example: "sonar-postgres-sink".
>   2. Associate the database_id to only one sink.

Routing logs of a single instance to a Pub/Sub Topic (One-to-One)

To send audit logs of a single instance to a single Pub/Sub topic, use the filter below while creating the Log Router Sink. Replace the "<GCP-project-id>" and <postgres-instance-name> with your own values in the filters.

```
resource.labels.database_id="<GCP-project-id>:<postgres-instance-name>" AND (logName="projec
```

Routing logs of multiple instances to a Pub/Sub Topic (Many-to-One)

To send audit logs of multiple instances to a single Pub/Sub topic, use the filter below while creating the Log Router Sink.

```
resource.type="cloudsql_database" AND (
resource.labels.database_id="<GCP-project-id>:<postgres-INSTANCE1-name>" OR
resource.labels.database_id="<GCP-project-id>:<postgres-INSTANCE2-name>")
AND
```

```
( logName=("projects/<GCP-project-id>/logs/cloudaudit.googleapis.com%2Fdata_access" OR "proj
```

Creating a Pub/Sub Subscription

Create a Pub/Sub subscription by following the instructions at Create pull subscriptions. For more information on Pub/Sub Topics and Subscriptions, see Overview of the Pub/Sub service and Create a topic.

> **Note:**
>
> - It is recommended to have one Agentless Gateway pull from a given Pub/Sub subscriptions. Multiple Agentless Gateways pulling from the same Pub/Sub subscription may result in the ingestion of partial data.
> - A single Pub/Sub subscription can store the audit logs of multiple instances of the same Server Type. However, the audit logs of different Server Types cannot be sent to the same Pub/Sub subscription.
> - If several instances of the same Server Type write audit data to a single Pub/Sub subscription, connecting any one of the data source assets or the Pub/Sub subscription asset will pull all the audit data from the subscription. The Pub/Sub asset will stay connected to the gateway and will pull all the audit data until all the data source assets using it are disconnected.

Network Prerequisites

Below is an overview of the necessary requirements to enable the Agentless Gateway to communicate with data sources and other components across the network. Please ensure these are completed by a Network Administrator.

Agentless Gateway Firewall Settings

DSF reads audit events from Google's Pub/Sub service running on the Google cloud. Ensure the Agentless Gateway host is able to access the Pub/Sub subscription. Note that:

- The Agentless Gateway host must be able to reach the Google Cloud Pub/Sub API endpoint `pubsub.googleapis.com`. If the gateway is configured with dynamic subscribers (a non-default setting), it must also reach `monitoring.googleapis.com`.
- The firewall needs to allow traffic from the Agentless Gateway on several outbound ports to enable communication with the Pub/Sub subscription:
  - Ports 80 and 443: for HTTP and HTTPS requests to the Pub/Sub subscription.
-
  - 5432: for connecting to the database to turn on audit policies and run SDM scans.

✔STEP 2: Enabling Audit on the Data Source

Once the required permissions and information have been obtained, please complete the following steps to enable audit on the data source.

Enable Auditing on Cloud SQL

The Cloud SQL service provides three distinct types of messages:

- System Event audit logs: Identifies automated Google Cloud actions that modify the configuration of resources.
- Admin Activity audit logs: Includes "admin write" operations that write metadata or configuration information.
- Data Access audit logs: Includes "admin read" operations that read metadata or configuration information. Also includes "data read" and "data write" operations that read or write user provided data.

System Event and Admin Activity audit logs are enabled by default. Data Access audit logs are disabled by default. To enable them, execute the following steps:

1. From the Google Cloud console, navigate to IAM & Admin > Audit Logs.
2. In the Data Access audit logs configuration table, select Cloud SQL service in the Service column.
3. In the Log Types tab that appears on the right, enable “Admin Read”, “Data Read”, and “Data Write”. For more information, see Enable Data Access audit logs.
4. Click Save.

For more information on Cloud SQL for PostgreSQL auditing, see Cloud SQL for PostgreSQL audit logging.

Enable Auditing on Cloud SQL for PostgreSQL instance

Auditing can be configured on the Cloud SQL for PostgreSQL instance by modifying the database flags. For more information, see Configuring database flags. For example, to enable audit on the instance perform the following steps from the Google Cloud console:

- Navigate to the left sidebar in Google Home console.
- Click on “SQL”.
- Click on your Cloud SQL for PostgreSQL instance.
- Click “Edit”.
- Under the “Configuration options” section, click “Flags”.

Set the following flags and click "Save" when finished. A text box will appear notifying you that your instance will need to be restarted, click "save and restart".

Flag	Value	Notes
cloudsql.enable_pgaudit	on	
log_error_verbosity	verbose	
log_connections	on	
log_disconnections	on	
log_hostname	on	
pgaudit.log	all	Setting the **pgaudit.log** parameter

		to **all** will log all events. To set up a more granular Audit Policy, see pgAudit documentation.

For DSF version 4.16 and newer set this additional flag:

Flag	Value	Notes
log_line_prefix	SONAR_AUDIT=1|TIMESTAMP=%m|APPLICATION_NAME=%a|USER=%u|DATABASE=%d|REMOTE_HOST_AND_PORT=%r|SQL_STATE=%e|SESSION_ID=%c|SESSION_START=%s|PROCESS_ID=[%p]|VIRTUAL_TRANSACTION_ID=%v|TRANSACTION_ID=%x|	It is required to use the log_line_prefix exactly as shown. Any modifications may cause system errors.

After enabling the database flags, run the `CREATE EXTENSION` command as shown below using any compatible PostgreSQL client.

```
CREATE EXTENSION pgaudit;
```

Enable Slow Query Auditing (Optional)

Slow query logs audit SQL statements that run longer than a specified duration. This duration is set using the flag "log_min_duration_statement" under the "**Configuration Options**" section in addition to the audit flags enabled in the previous step.

Flag	Value	Notes
log_min_duration_statement	<value>	<value> is your desired threshold in milliseconds and may be updated as desired

Note:

- Turning on Slow Query Monitoring will not interfere with standard audit collection.
- Queries that exceed the minimum duration threshold will be mapped to a separate collection called sonargd.slow_query.
- Make sure the **log_statement** parameter is set to **None**. Any modifications may cause slow query logs to not be ingested.
- This step is **optional** and required only if you choose to audit slow query logs.
- The following setting should be done IN ADDITION to all previous steps.

✔STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

GCP PostgreSQL Asset Specifications

GCP Pub Sub Asset Specifications

GCP Asset Specifications

> **Note:**
>
> The audit logs of multiple PostgreSQL instances can be routed to one Pub/Sub topic. If you have such a setup then:
>
> 1. Create separate assets for each PostgreSQL instance. Only one asset is required for the Pub/Sub subscription asset.
> 2. Connecting the Pub/Sub asset will pull the audit data for all the servers. The Pub/Sub asset stays connected and pulls all the audit data until all the instance assets are disconnected.

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.

2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.

3. In the Data Source Type dropdown menu, select a data source.

4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.

5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click Advanced.

6. Under Monitoring, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.

7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the Adding Assets via Unified Settings Console (USC) documentation.

Discover Assets using Cloud Accounts in USC

The Cloud Accounts tab allows you to run the Discovery playbook at any time to automatically discover and onboard new assets. Follow the instructions below to add a new cloud account and run Discovery via USC.

To add a new cloud account:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Cloud Accounts tab to view the cloud accounts of the DSF Hub. Click **Add** to open the Add Cloud Account form.
3. In the Cloud Account Type dropdown menu, select Amazon, Azure, or Google Cloud.
4. Specific configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. Under Connection, select the authentication mechanism that the Agentless Gateway will use to connect to the cloud account.
   - For Amazon: in the Auth Mechanism dropdown menu, select Profile, Key, IAM Role or Default.
   - For Azure: in the Auth Mechanism dropdown menu, select Client Secret, Auth File, or Managed Identity.
   - For Google Cloud: in the Auth Mechanism dropdown menu, select Default or Service Account.
6. Depending on the selected cloud account type and authentication mechanism, different fields will be displayed under the Connection section. Fill them out accordingly.
7. Under Monitoring, select the Agentless Gateway that will own the cloud account in the Gateway field, then click **Save**.

To run Discovery in USC:

1. Locate the desired cloud account asset in USC as described above.
2. Click on the kebab menu icon to the right of the cloud account on which you want to run the Discovery feature.
3. Click **Start Discovery**. Discovery will start running in the background.
4. After Discovery has run successfully, locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on importing data source assets using cloud accounts in USC, see the Discovering Data Sources in DSF Hub Cloud Accounts using USC documentation.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

1. From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specification documentation as a guide for filling in the necessary field values.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.

5. To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Management** > **Audit Actions** > **Connect Gateway** to start collecting audit data.

For more information, please see the Adding Assets via the Import Assets Page documentation.

> **Note:**
>
> - Asset template spreadsheet name for Cloud SQL for PostgreSQL: **GOOGLECLOUD_CLOUDSQL_POSTGRESQL_template.xlsx**
> - If you are sending the audit logs of multiple PostgreSQL instances to a single Pub/Sub, filter for the assets and click on the "Management" button displayed above the assets → Audit Actions → Connect/Disconnect Gateway to connect or disconnect all assets.

Discover Assets using Cloud Accounts in the Assets Dashboard

> **Note:**
>
> This section can be skipped if the data source asset(s) have already been onboarded using spreadsheets in the previous section.

After configuring and onboarding a cloud account asset with the necessary permissions, the Asset Discovery feature can find and onboard related data source assets with just a few clicks.

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download.**
3. Use the Asset Specifications documentation as a guide for completing the asset template spreadsheet for the cloud account asset.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **Validate All**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the cloud account asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Discovery**. Choose the data source type to be discovered to start the Discovery playbook.
9. After the Discovery playbook has run successfully, locate the desired data source asset.
10. Click the kebab menu to the right of the asset, then click **Management** > **Audit Actions** > **Connect Gateway** to start collecting audit data.

For more information on discovering assets, please see the Adding Assets via Asset Discovery documentation.

> **Note:**
>
> Asset template name for Google Cloud Account Asset; choose any ONE of the templates listed below based on the type of authentication mechanism:
>
> - **GOOGLECLOUD_SERVICE_ACCOUNT_template.xlsx**
> - **GOOGLECLOUD_DEFAULT_template.xlsx**

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see Using DSF Open APIs.

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via https://supportportal.thalesgroup.com/csm.

If you encounter unexpected issues, review the services and log files listed below on the Agentless Gateway host(s) to identify the root cause.

DSF Hub versions 15.3 and newer :

- Gateway log file: `$JSONAR_LOGDIR/gateway/syslog/postgresql_pubsub.log`
- Gateway pull log file: $`JSONAR_LOGDIR/gateway/pull/google-pubsub/gateway-pull.log`
- Run the following command to verify that the required services are active:

```
systemctl status -l gateway-pull@google-pubsub.service
systemctl status -l sonarrsyslog
```

DSF Hub versions 4.11 to 15.2 :

- Gateway log file: `JSONAR_LOGDIR/gateway/syslog/postgresql_pubsub.log`
- SonarGD log file: `JSONAR_LOGDIR/sonargd/sonargd.log`
- Run the following command to verify that the required services are active

```
systemctl status -l sonargd
systemctl status -l sonarrsyslog
```

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via https://supportportal.thalesgroup.com/csm. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

- Using DSF Open APIs: A guide to onboarding and managing assets via DSF Open APIs.
- Adding assets via Unified Settings Console (USC): Instructions to add and edit data source assets via the USC.
- Adding Assets via the Import Assets Page: Instructions to add data source assets via the Asset Dashboard.
- Using the Data Security Fabric Portal: An overview of the Data Security Fabric (DSF) portal.
- Data Security Coverage Tool: Systems and version compatibility with DSF Hub.

Document generated by Confluence on Jun 19, 2026 01:21

Atlassian

Cloud SQL for PostgreSQL Performance Improvements for v 4.16+

This topic reviews the steps required to audit activity on GCP PostgreSQL databases and send that information to DSF Hub for auditing, analysis, detecting and preventing security events.

Introduction

Starting in DSF version 4.16, customers monitoring Google Cloud SQL for PostgreSQL are offered the option to improve the consumption rate pulling data from Google Cloud Pub/Sub and the overall data richness of the audit documents by enabling multi-processing.

Overview

The process involves two steps following the upgrade:

1. Add a `log_line_prefix` flag to the Cloud SQL for PostgreSQL server. This change is recommended for audit documents to display richer data more consistently, regardless of the consumption rate improvement.
2. Update the asset document to include the “subscribers” field, to determine the number of pull processes that will be used. This field expects a positive integer value (up to 16) or the string “dynamic”, as per the GCP Pub Sub Asset Specifications.

Detailed Procedure

Step 1: Add a `log_line_prefix` database flag

1. Navigate to your Cloud SQL for PostgreSQL server’s page.

2. Click “EDIT”.

3. Expand the “Flags” section.

4. Click “Add a database flag”.

5. From the drop down, choose “log_line_prefix”.

6. Add the following value exactly as shown below. Any modifications to this format may result in system errors.

```
SONAR_AUDIT=1|TIMESTAMP=%m|APPLICATION_NAME=%a|USER=%u|DATABASE=%d|REMOTE_HOST_AND_PORT=%
```

7. Click “DONE”.

8. Changes should take effect immediately. To validate this, navigate to Google Cloud’s Log Explorer. Filter for your Cloud SQL for PostgreSQL server; audit logs should now display the `log_line_prefix` in the "textPayload" field, starting with `SONAR_AUDIT=1|....`.

```
logName: "projects/cloudsql-postgres-audit/logs/cloudsql.googleapis.com%2Fpostgres.log"
receiveTimestamp: "2024-05-16T04:39:50.227759892Z"
resource: {2}
severity: "INFO"
textPayload: "SONAR_AUDIT=1|TIMESTAMP=2024-05-16 04:39:48.117 UTC|APPLICATION_NAME=[unknown]
timestamp: "2024-05-16T04:39:48.117600Z"
}
```

Step 2: Update your Asset Document

1. Login to DSF Hub and navigate to the ”Asset Dashboard”.

2. From the navigation menu, select “Data Activity Monitoring” → “Search & Analyze” → “Dashboard”.

3. Locate your Cloud SQL for PostgreSQL asset, and from the side menu choose “Update Document”.

4. Scroll to the bottom, and add an “Outside metadata” field called “subscribers”.

5. Assign this field a positive integer value between 1-16 or the string “dynamic”.

6. Click “Add field”.

7. Click “Update”.

8. The new field will be synced to the Agentless Gateway that owns the asset within 10 minutes, or manually trigger the sync job by clicking “Management” → “Sync with Agentless Gateways”.

On your Agentless Gateway, you can monitor the sonarGD log to verify the desired number of subscribers are open and pulling:

```
$ sudo - su
```

```
$ source /etc/sysconfig/jsonar
$ tail -f $JSONAR_LOGDIR/sonargd/sonargd.log

I 2024-05-16 19:15:00.463 remote_1   PUBSUB - projects/cloudsql-postgres-audit/subscriptions
I 2024-05-16 19:15:19.509 remote_1   PUBSUB - projects/cloudsql-postgres-audit/subscriptions
I 2024-05-16 19:15:19.509 remote_1   PUBSUB - projects/cloudsql-postgres-audit/subscriptions
I 2024-05-16 19:15:19.509 remote_1   PUBSUB - projects/cloudsql-postgres-audit/subscriptions
I 2024-05-16 19:15:19.509 remote_1   PUBSUB - projects/cloudsql-postgres-audit/subscriptions
I 2024-05-16 19:15:19.509 remote_1   PUBSUB - projects/cloudsql-postgres-audit/subscriptions
I 2024-05-16 19:15:19.509 remote_1   PUBSUB - projects/cloudsql-postgres-audit/subscriptions
```

Monitor the respective gateway service log to follow the number of documents being generated:

```
$ sudo tail -f ${JSONAR_LOGDIR}/gateway/syslog/postgresql_pubsub.log

2024-05-17 17:41:23,957 INFO Spool 3 bsons to batch file /data/jsonar/data/gateway/4/spool/s
2024-05-17 17:45:23,958 INFO Spool 777 bsons to batch file /data/jsonar/data/gateway/4/spool
2024-05-17 17:45:23,958 INFO Spool 6 bsons to batch file /data/jsonar/data/gateway/4/spool/s
2024-05-17 17:45:23,958 INFO Spool 777 bsons to batch file /data/jsonar/data/gateway/4/spool
```

Document generated by Confluence on Jun 19, 2026 01:21

Atlassian

Cloud SQL for SQL Server Data Sources

There are two available methods for auditing Cloud SQL for SQL Server. The recommended approach is to use a Storage Bucket, which leverages the native SQL Server Audit feature and captures all server-level audit action groups supported by SQL Server Audit. Alternatively, the Pub/Sub method audits only System Events, Admin Activity, and Data Access logs available on GCP, offering a more limited view of server activity.

Please select a link below to view the step-by-step instructions for each auditing method:

- Cloud SQL for SQL Server with PubSub Onboarding Steps
- Cloud SQL for SQL Server with Storage Bucket Onboarding Steps

Document generated by Confluence on Jun 19, 2026 01:21

Atlassian

Cloud SQL for SQL Server with PubSub Onboarding Steps

This topic reviews the general steps required to audit activity on Cloud SQL for SQL Server with Pub Sub databases and send that information to DSF Hub for auditing, analysis, detecting and preventing security events.

> **Note:**
>
> The recommended method to audit a Cloud SQL for SQL Server instance is by using a Storage Bucket. For

detailed instructions, see Cloud SQL for SQL Server with Storage Bucket Onboarding Steps.

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The following main topics are included:

- An overview of the available scope for the data source, given its native audit data.
- A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.
- Instructions on how to enable audit on the data source and collect data using DSF Hub.
- Initial troubleshooting steps and technical support information.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps. Please click on each of the steps below to display the content.

STEP 1: Onboarding Prerequisites

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

Permissions for Google Cloud Account Principal

Grant the following IAM roles to a cloud account principal:

- **Logging Admin** and **Project IAM Admin** roles: for enabling Data Access logs, viewing logs and creating a Log Router Sink.
- **Pub/Sub Admin** role: for creating a Pub/Sub Topic and Subscription.
- **CloudSQL Admin** role: for modifying database-specific flags.
- **Service Account Admin** and optionally **Service Account Key Admin** roles: for creating a service account and generating keys (if using the "service_account" authentication mechanism). See Permissions and roles for more information on the list of individual permission contained in each predefined role.

Creating a Google Cloud Service Account

DSF supports the following two authentication mechanisms for accessing Google Cloud resources. Note that both authentication mechanisms make use of a Service Account. The process for creating a Service Account is described in Creating and managing service accounts.

- **Service Account:** to use the `service_account` authentication mechanism, a JSON-formatted key file needs to be generated holding the credentials that the Agentless Gateway will use to access audit logs from the Pub/Sub subscription. Ensure that the key file is copied to the Agentless Gateway host and it is owned by the `sonarw` user. For more information on how to create a service account key, see Creating and managing service account keys.

- **Default:** to use the `default` authentication mechanism, attach the Service Account created above to the Agentless Gateway hosted on a Google Cloud Compute Engine virtual machine instance. For more information on how to attach a Service Account to a GCP Compute Engine virtual machine, see Change the attached service account.

The Service Account should be granted the following roles:

- Pub/Sub Subscriber
- Pub/Sub Viewer
- Viewer

> **Note:**
>
> It is recommended to keep both the Service Account and the Agentless Gateway Google Cloud VM instance within the same Google Cloud Project.

Google Cloud Log Routing Overview: Log Router Sinks, Pub/Sub Topics and Subscriptions

Log Router Sinks control how Logging routes logs in Google Cloud. Using sinks, you can route audit logs to one or more supported destinations. Please follow the instructions in Route logs to supported destinations. The following summarizes the steps to be completed:

1. Create a Log Router sink that sends the audit logs to a new or existing Pub/Sub Topic as the destination service.

    a. In the "Choose logs to include in sink" panel, filter for your data source’s audit logs. Follow the below examples to create a filter for your data source configuration.

2. Create a Pub/Sub Subscription that receives messages from the Pub/Sub Topic. The Agentless Gateway service will pull the audit logs from this subscription.

The gcloud command line tool - linked to a project owner service account - may also be used to create the Log Router Sink and export the audit logs to a Pub/Sub Topic.

Routing logs of a Cloud SQL for SQL Server instance to a Pub/Sub Topic

Use the filter below while creating the Log Router Sink to send the SQL Server audit logs to a single Pub/Sub topic, replacing the <GCP-project-id> and <mssql-instance-name> with your own values.

```
resource.type="cloudsql_database"
resource.labels.database_id="<GCP-project-id>:<mssql-instance-name>"
```

> **Note:**
>
> It is recommended to not have more than one Log Router Sink with the same database_id value and to start the Sink name with the word "sonar".

Creating a Pub/Sub Subscription

Create a Pub/Sub subscription by following the instructions at Create pull subscriptions. For more information on Pub/Sub Topics and Subscriptions, see Overview of the Pub/Sub service and Create a topic.

> **Note:**
>
> - It is recommended to have one Agentless Gateway pull from a given Pub/Sub subscriptions. Multiple Agentless Gateways pulling from the same Pub/Sub subscription may result in the ingestion of partial data.
> - A single Pub/Sub subscription can store the audit logs of multiple instances of the same Server Type. However, the audit logs of different Server Types cannot be sent to the same Pub/Sub subscription.
> - If several instances of the same Server Type write audit data to a single Pub/Sub subscription, connecting any one of the data source assets or the Pub/Sub subscription asset will pull all the audit data from the subscription. The Pub/Sub asset will stay connected to the gateway and will pull all the audit data until all the data source assets using it are disconnected.

Network Prerequisites

Below is an overview of the necessary requirements to enable the Agentless Gateway to communicate with data sources and other components across the network. Please ensure these are completed by a Network Administrator.

Agentless Gateway Firewall Settings

DSF reads audit events from Google's Pub/Sub service running on the Google cloud. Ensure the Agentless Gateway host is able to access the Pub/Sub subscription. Note that:

- The Agentless Gateway host must be able to reach the Google Cloud Pub/Sub API endpoint `pubsub.googleapis.com`. If the gateway is configured with dynamic subscribers (a non-default setting), it must also reach `monitoring.googleapis.com`.
- The firewall needs to allow traffic from the Agentless Gateway on several outbound ports to enable communication with the Pub/Sub subscription:
  - Ports 80 and 443: for HTTP and HTTPS requests to the Pub/Sub subscription.

    443 - For Discovery API calls

Add Agentless Gateway IP to Allowlist

> **Note:**
>
> This step is required only if you want to "Test Connection" to a database or run SDM scans.

To allow external connections to the database, you must set up authorization for database connections. Please refer to the Enabling public IP and adding an authorized address or address range documentation for more details.

STEP 2: Enabling Audit on the Data Source

Once the required permissions and information have been obtained, please complete the following steps to enable audit on the data source.

The Cloud SQL service provides three distinct types of messages:

- System Event audit logs: these messages identify automated Google Cloud actions that modify the configuration of resources.
- Admin Activity audit logs: these messages include "admin write" operations that write metadata or configuration information.
- Data Access audit logs: these messages include "admin read" operations that read metadata or configuration information. They also include "data read" and "data write" operations that read or write user provided data.

System Event and Admin Activity audit logs are enabled by default. Data Access audit logs are disabled by default. To enable them, complete the following steps:

1. From the Google Cloud console, navigate to IAM & Admin > Audit Logs.
2. In the Data Access audit logs configuration table, select Cloud SQL service in the Service column.
3. In the Log Types tab that appears on the right, enable “Admin Read”, “Data Read”, and “Data Write”. For more information, see Available audit logs.
4. Click Save.

STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

GCP MS SQL Server Asset Specifications

GCP Pub Sub Asset Specifications

GCP Asset Specifications

> **Note:**
>
> Audit logs of multiple SQL Server instances can be routed to one Pub/Sub Topic by following these steps:
>
> 1. Create separate assets for each SQL Server instance. Only one asset is required for the Pub/Sub subscription.
> 2. Connect the Pub/Sub asset to pull the audit data for all the instances.
>
> The Pub/Sub asset will stay connected and pull all of the audit data until all of the instance assets are disconnected.

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.

2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.

3. In the Data Source Type dropdown menu, select a data source.

4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.

5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click Advanced.

6. Under Monitoring, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.

7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your

asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the Adding Assets via Unified Settings Console (USC) documentation.

> **Note:**
>
> While creating the assets in USC, select the "MsSQL" Audit Type.

Discover Assets using Cloud Accounts in USC

The Cloud Accounts tab allows you to run the Discovery playbook at any time to automatically discover and onboard new assets. Follow the instructions below to add a new cloud account and run Discovery via USC.

To add a new cloud account:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Cloud Accounts tab to view the cloud accounts of the DSF Hub. Click **Add** to open the Add Cloud Account form.
3. In the Cloud Account Type dropdown menu, select Amazon, Azure, or Google Cloud.
4. Specific configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. Under Connection, select the authentication mechanism that the Agentless Gateway will use to connect to the cloud account.
   - For Amazon: in the Auth Mechanism dropdown menu, select Profile, Key, IAM Role or Default.
   - For Azure: in the Auth Mechanism dropdown menu, select Client Secret, Auth File, or Managed Identity.
   - For Google Cloud: in the Auth Mechanism dropdown menu, select Default or Service Account.
6. Depending on the selected cloud account type and authentication mechanism, different fields will be displayed under the Connection section. Fill them out accordingly.
7. Under Monitoring, select the Agentless Gateway that will own the cloud account in the Gateway field, then click **Save**.

To run Discovery in USC:

1. Locate the desired cloud account asset in USC as described above.
2. Click on the kebab menu icon to the right of the cloud account on which you want to run the Discovery feature.
3. Click **Start Discovery**. Discovery will start running in the background.
4. After Discovery has run successfully, locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on importing data source assets using cloud accounts in USC, see the Discovering Data Sources in DSF Hub Cloud Accounts using USC documentation.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

1. From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specification documentation as a guide for filling in the necessary field values.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Management** > **Audit Actions** > **Connect Gateway** to start collecting audit data.

For more information, please see the Adding Assets via the Import Assets Page documentation.

Discover Assets using Cloud Accounts in the Assets Dashboard

> **Note:**
>
> This section can be skipped if the data source asset(s) have already been onboarded using spreadsheets in the previous section.

After configuring and onboarding a cloud account asset with the necessary permissions, the Asset Discovery feature can find and onboard related data source assets with just a few clicks.

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download.**
3. Use the Asset Specifications documentation as a guide for completing the asset template spreadsheet for the cloud account asset.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **Validate All**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the cloud account asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Discovery**. Choose the data source type to be discovered to start the Discovery playbook.
9. After the Discovery playbook has run successfully, locate the desired data source asset.
10. Click the kebab menu to the right of the asset, then click **Management** > **Audit Actions** > **Connect Gateway** to start collecting audit data.

For more information on discovering assets, please see the Adding Assets via Asset Discovery documentation.

Note:

Asset template name for Cloud SQL for SQL Server: ***GOOGLECLOUD_CLOUDSQL_SQLSERVER_template.xlsx***

Asset template name for Google Cloud Account asset; choose any one of the templates listed below based on the type of authentication mechanism:

- **GOOGLECLOUD_SERVICE_ACCOUNT_template.xlsx**
- **GOOGLECLOUD_DEFAULT_template.xlsx**

When the "Connect Gateway" playbook is run, select the "MSSQL" audit_type.

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see Using DSF Open APIs.

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via https://supportportal.thalesgroup.com/csm.

DSF Hub versions 15.3 and newer :

- Gateway log file: $JSONAR_LOGDIR/gateway/syslog/mssql_pubsub.log
- Gateway pull log file: $JSONAR_LOGDIR/gateway/pull/google-pubsub/gateway-pull.log

- Run the following command to verify that the required services are active:

```
systemctl status -l gateway-pull@google-pubsub.service
systemctl status -l sonarrsyslog
```

DSF versions 4.11 to 15.2:

- Gateway log file: $JSONAR_LOGDIR/gateway/syslog/mssql_pubsub.log
- SonarGD log file : $JSONAR_LOGDIR/sonargd/sonargd.log

- As the "root" user, run the following command to verify the status of the services:

```
systemctl status -l sonargd
systemctl status -l sonarrsyslog
```

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via https://supportportal.thalesgroup.com/csm. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

- Using DSF Open APIs: A guide to onboarding and managing assets via DSF Open APIs.
- Adding assets via Unified Settings Console (USC): Instructions to add and edit data source assets via the USC.
- Adding Assets via the Import Assets Page: Instructions to add data source assets via the Asset Dashboard.
- Using the Data Security Fabric Portal: An overview of the Data Security Fabric (DSF) portal.
- Data Security Coverage Tool: Systems and version compatibility with DSF Hub.

Document generated by Confluence on Jun 19, 2026 01:21

Atlassian

Cloud SQL for SQL Server with Storage Bucket Onboarding Steps

This topic reviews the steps required to audit activity on GCP SQL Server with storage bucket databases and send that information to DSF Hub for auditing, analysis, detecting and preventing security events.

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The following main topics are included:

- An overview of the available scope for the data source, given its native audit data.
- A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.
- Instructions on how to enable audit on the data source and collect data using DSF Hub.
- Initial troubleshooting steps and technical support information.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps. Please click on each of the steps below to display the content.

STEP 1: Onboarding Prerequisites

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

IT Prerequisites

Below is an overview of the necessary steps which will need to be completed on the Agentless Gateway host or the data source. Please ensure these steps are completed by an IT Administrator.

- You have sudo privileges on the Agentless Gateway host to copy and edit the sonargd.conf file for changing the default retention policy mechanism if desired.

There are three available retention policy mechanisms for handling the audit log files once they are downloaded from the Cloud Storage Bucket onto the Agentless Gateway.

1. The "delete" mechanism: the audit log files are deleted from the Storage Bucket. This is enabled by default.
2. The "archive" mechanism: the audit log files are moved to a subfolder within the Storage Bucket called "imperva-archive".
3. The "marker" mechanism: each time an audit log file is downloaded, a marker is either created at the "now" timestamp minus the "lookback" (one day, by default) for the first audit pull, or fetched from an existing marker. The lookback value can be found in the sonargd configuration file under gcp-cloud-storage.lookback. The marker is saved in the lmrm__scheduler.lmrm__last_run collection and is unique per bucket asset_id. The audit log files are sorted from oldest to newest before the download begins. After each audit file downloads, the marker is updated to the time_created of the audit file. The audit files are not moved or altered in any way on the bucket itself.

Note:

The "archive" and "marker" mechanisms are available from DSF version 4.13 onward. For versions prior to 4.13, the only available retention policy mechanism is "delete".

Each retention policy mechanism uses a "delay" of 15 minutes when it downloads the files from the Storage Bucket. This is to avoid downloading files that did not reach the MAXSIZE set when the "server audit" was created, as incomplete files are sometimes uploaded to the bucket by GCP. The delay value can be found in the sonargd configuration file under gcp-cloud-storage.delay.

By default, the retention policy mechanism is "delete". To change it, connect to your Agentless Gateway host and complete the following steps as the "root" user:

1. Source the DSF variables.

```
source /etc/sysconfig/jsonar
```

2. Copy the "sonargd.conf" file from the JSONAR_BASEDIR to the JSONAR_LOCALDIR directory.

```
cp -n $JSONAR_BASEDIR/etc/sonar/sonargd.conf $JSONAR_LOCALDIR/sonargd/sonargd.conf
```

3. In "$JSONAR_LOCALDIR/sonargd/sonargd.conf" find the below lines and change the value of "mechanism" to one of: "archive", "marker", or "delete".

```
gcp-cloud-storage:
    lookback: P1D
    delay: PT15M
    mechanism: delete
```

4. Restart the SonarGD service.

```
systemctl restart sonargd
```

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

Permissions for Google Cloud Account Principal

Grant the following permissions/IAM roles to a cloud account principal:

- CloudSQL Admin role: for enabling the "Sql Server audit" option on your Cloud SQL for SQL Server instance and for creating an IAM Role with the privileges to the audit the Cloud Storage Bucket.
- storage.buckets.create privilege: for creating a Cloud Storage Bucket and setting it as the destination for the SQL Server audit log files.
- Service Account Admin (and optionally Service Account Key Admin) role(s): for creating a service account and generating service account keys if using the "service_account" authentication mechanism on the GCP Cloud Account asset. See Permissions and roles for more information on the list of individual permission contained in each predefined role.

Creating a Google Cloud Service Account

DSF supports the following two authentication mechanisms for accessing Google Cloud resources. Note that both authentication mechanisms make use of a Service Account. The process for creating a Service Account is described in Creating and managing service accounts.

- **Service Account:** to use the “service_account” authentication mechanism, a JSON-formatted key file needs to be generated holding the credentials that the Agentless Gateway will use to access audit logs from the GCP Cloud Storage Bucket. Ensure that the key file is copied to the Agentless Gateway host and it is owned by the "sonarw" user. For more information on how to create a service account key, see Creating and managing service account keys.

- **Default:** to use the "default" authentication mechanism, attach the Service Account created above to the Agentless Gateway hosted on a Google Cloud Compute Engine virtual machine instance. For more information on how to attach a Service Account to a GCP Compute Engine virtual machine, see Change the attached service account.

The Service Account should be granted the following permissions for the respective retention policy mechanism so that the Agentless Gateway can access the audit log files in the Cloud Storage Bucket:

Retention Policy Mechanism	Necessary Permissions
Delete	▪ storage.objects.get ▪ storage.objects.list ▪ storage.buckets.get ▪ storage.objects.delete
Marker	▪ storage.buckets.get ▪ storage.objects.get ▪ storage.objects.list
Archive	▪ storage.objects.get ▪ storage.objects.list ▪ storage.buckets.get ▪ storage.objects.delete ▪ storage.objects.create

Note:

- Ensure that the permissions added to the Service Account match the retention policy mechanism set in "sonargd.conf". Mismatched configurations may lead to duplicated data consumption on the DSF Hub.
- It is recommended to keep both the Service Account and the Agentless Gateway Google Cloud VM instance within the same Google Cloud Project.

Network Prerequisites

Below is an overview of the necessary requirements to enable the Agentless Gateway to communicate with data sources and other components across the network. Please ensure these are completed by a Network Administrator.

Ensure the following ports are open on the Agentless Gateway host:

1433: for connecting to the database for turning on audit policies and running SDM scans.

443: for discovery API calls.

Database Prerequisites

Below are the requirements for granting database permissions for audit log retrieval. Please ensure these are completed by a Database Administrator.

To create a server audit, you will need to connect as the default "sqlserver" user. Please obtain the credentials of this user from the DBA.

Validate Connection to Database

1. To verify the connection to the SQL Server master database as the "sqlserver" user, login as the "sonarw" user on the Agentless Gateway host:

```
sudo su - sonarw
```

2. Source the DSF variables:

```
source /etc/sysconfig/jsonar
```

3. Connect to the master database:

```
${JSONAR_BASEDIR}/bin/tsql -S <SQL Server instance IP> -p <port> -D master -U sqlserver -
```

Address any errors here before continuing.

✔STEP 2: Enabling Audit on the Data Source

Once the required permissions and information have been obtained, please complete the following steps to enable audit on the data source.

For each SQL Server instance, to enable auditing on an instance, edit its "Flag and parameters" section:

1. In the Google Cloud console, go to the Cloud SQL Instances page.
2. To open the Overview page of an instance, click the instance name.
3. Click Edit.
4. In the "Customize your instance" section, click "Flags and parameters".
5. Check the checkbox beside "Enable SQL Server audit".
6. Under "Choose where to store audit logs", provide the path of a folder within your Cloud Storage Bucket with the same name as that of your SQL Server instance's "region".
   For example, if your SQL Server instance is in "us-east1", then the Cloud Storage location should look like this:

```
<bucket_name>/us-east1/
```

> **Note:**
> - For all SQL Server instances, ensure that the Cloud Storage location (Bucket folder) selected uses the same name as the SQL Server instance region.

7. Click Advanced Options.
8. In the "Choose how many days of logs to retain" option, select a period of at least 24 hours. This is to ensure that DSF can retrieve older logs if the SQL Server instance or the Agentless Gateway is inaccessible for a period of time.
9. Click Save to apply your changes.
10. Connect to your SQL Server instance and create a server audit policy. To do this, follow the instructions in Step 4 "Managing Audit Policies".

✔STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

GCP MS SQL Server Asset Specifications

GCP Cloud Storage Bucket Asset Specifications

GCP Asset Specifications

> **Note:**
> - It is recommended that one Agentless Gateway pull from one given storage bucket. Multiple gateways pulling from the same bucket may result in the ingestion of partial or no data.
> - The audit logs of multiple SQL Server instances may be sent to one storage bucket. If you have such a setup:
>   - Create separate assets for each SQL Server instance. Only one asset is required for the bucket asset.
>   - Connecting the bucket asset will pull the audit data for all the servers. If one of the SQL Server assets is disabled, data from that server will still be read but certain field values that rely on the asset may be

> missing from the documents. The Storage Bucket Aggregator asset reads all the audit data until all the SQL Server assets are disconnected.

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.
3. In the Data Source Type dropdown menu, select a data source.
4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click Advanced.
6. Under Monitoring, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.
7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the Adding Assets via Unified Settings Console (USC) documentation.

> **Note:**
>
> While creating the assets in USC, select the "Bucket" Audit Type.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

1. From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.

3. Use the Asset Specification documentation as a guide for filling in the necessary field values.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Management** > **Audit Actions** > **Connect Gateway** to start collecting audit data.

For more information, please see the Adding Assets via the Import Assets Page documentation.

> **Note:**
>
> - To import the Cloud SQL for SQL Server asset, at least three assets are required: a GCP account asset, a Cloud SQL Storage Bucket asset and at least one Cloud SQL for SQL Server Database asset.
>
> - Asset template name for Cloud SQL for SQL Server with Storage Bucket: **GOOGLECLOUD_CLOUDSQL_SQLSERVER_WITH_STORAGE_BUCKET_template.xlsx**
> - Asset template name for Google Cloud Account Asset; choose any ONE of the templates listed below based on the type of authentication mechanism:
>   - **GOOGLECLOUD_SERVICE_ACCOUNT_template.xlsx**
>   - **GOOGLECLOUD_DEFAULT_template.xlsx**
> - When the "Connect Gateway" playbook is run, select the "BUCKET" audit_type.

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see Using DSF Open APIs.

STEP 4: Managing Audit Policies

Enabling an Audit Policy via the DSF Hub is not available. Please create a server audit manually as shown below.

Connect with the default "sqlserver" user and run the following commands on the master database.

1. **Create a "server audit"**
   To set up the server audit, provide a unique identifier AUDIT_GUID. Please refer to the following documentation to determine the desired value: Create Server Audit.
   In the command below, replace the below parameters with your values:

   a. AUDIT_GUID - unique value

   b. <audit_name> - name of the server audit

```
USE master;
GO

CREATE SERVER AUDIT <audit_name> TO FILE (FILEPATH = '/var/opt/mssql/audit', MAXSIZE =
GO
```

2. **Enable the "server audit"**

```
ALTER SERVER AUDIT <audit_name> WITH (STATE = ON);
GO
```

3. **Create a Server Audit Specification** and add all the desired audit options. The example below covers all of the options for SQL Server 2019. For more details and examples for other SQL Server versions, see CREATE SERVER AUDIT SPECIFICATION.

   Replace the below parameters with your values:

   a. <audit_specification_name> - name of the audit specification

   b. <audit_name> - name of the server audit

```
CREATE SERVER AUDIT SPECIFICATION <audit_specification_name> FOR SERVER AUDIT <audit_na
GO
```

4. Verify that the Server Audit and Server Audit specification are active. The first command should return the field "status_desc: STARTED", and the second command should return the field "is_state_enabled: 1".

```
SELECT * FROM sys.dm_server_audit_status;
GO
SELECT * FROM sys.server_audit_specifications;
GO
```

5. To verify the audit log file path, run :

```
SELECT name, type_desc, log_file_path FROM sys.server_file_audits;
GO
```

   Once traffic is generated on the database, run the below command to check that audit files are being generated:

```
SELECT top 2 * FROM msdb.dbo.gcloudsql_fn_get_audit_file('/var/opt/mssql/audit/*', NULL,
```

```
GO
```

Note:

If more than one policy is enabled on the SQL Server instance, you may see duplicate events in the log.

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via https://supportportal.thalesgroup.com/csm.

On the Agentless Gateway host(s) please review the following:

1. Service log files:
   a. sonarrsyslog log file: $JSONAR_LOGDIR/gateway/syslog/mssql_xel.log
   b. sonargd log file: $JSONAR_LOGDIR/sonargd/sonargd.log

2. Run the following commands to verify the status of the gateway services:

```
systemctl status -l sonarrsyslog
systemctl status -l sonargd
```

3. After the audit policy is turned on and the database starts generating traffic:
   a. On the Google Cloud, check that audit log files are generated in the corresponding Google Cloud Storage Bucket folder /bucket_name/region
   b. On the Agentless Gateway host(s), check the $JSONAR_DATADIR/sonargd/incoming folder to see that audit files are being received. After a file is processed, it will be moved to $JSONAR_DATADIR/sonargd/archive

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via https://supportportal.thalesgroup.com/csm. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

- Using DSF Open APIs: A guide to onboarding and managing assets via DSF Open APIs.
- Adding assets via Unified Settings Console (USC): Instructions to add and edit data source assets via the USC.
- Adding Assets via the Import Assets Page: Instructions to add data source assets via the Asset Dashboard.
- Using the Data Security Fabric Portal: An overview of the Data Security Fabric (DSF) portal.
- Data Security Coverage Tool: Systems and version compatibility with DSF Hub.

Document generated by Confluence on Jun 19, 2026 01:21

Atlassian

Firestore Onboarding Steps

This topic reviews the steps required to audit activity on Firestoredatabases and send that information to DSF Hub for auditing, analysis, detecting and preventing security events.

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The following main topics are included:

- An overview of the available scope for the data source, given its native audit data.
- A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.
- Instructions on how to enable audit on the data source and collect data using DSF Hub.
- Initial troubleshooting steps and technical support information.

Audit Data Scope

The scope of audit data collection varies between data sources. Prior to enabling auditing, read through the auditing scope for this data source to identify any potential benefits or constraints. For more information on data collections, see Main Collections in sonargd Database.

- Firestore does not support session events. Audit events are collected only in the `instance` and `exception` collections.
- The `asset_id` field displayed in the asset dashboard is a placeholder. The actual `asset_id` values in the audit logs accurately represent the originating Firestore database.
- When navigating between GUI pages, Firestore repeatedly creates and terminates real-time listeners. Each time a listener is terminated, Firestore generates a **Listen** log event. These events appear as errors in the audit logs and are mapped to the `exception` collection.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps. Please click on each of the steps below to display the content.

STEP 1: Onboarding Prerequisites

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

Permissions for Google Cloud Account Principal

Grant the following IAM roles to a cloud account principal:

- **Logging Admin** and **Project IAM Admin**: Enables Data Access logs, viewing logs and creating a Log Router Sink.
- **Pub/Sub Admin**: Grants permissions to create a Pub/Sub Topic and Subscription.
- **Service Account Admin** (and optionally **Service Account Key Admin**): Allows the creation of a service account and generates keys when using the "service_account" authentication method. For more information on the permissions included in each role, see Permissions and roles.

Creating a Google Cloud Service Account

DSF supports the following two authentication mechanisms for accessing Google Cloud resources. Note that both authentication mechanisms make use of a Service Account. The process for creating a Service Account is described in Creating and managing service accounts.

- **Service Account:** to use the `service_account` authentication mechanism, a JSON-formatted key file needs to be generated holding the credentials that the Agentless Gateway will use to access audit logs from the Pub/Sub subscription. Ensure that the key file is copied to the Agentless Gateway host and it is owned by the `sonarw` user. For more information on how to create a service account key, see Creating and managing service account keys.
- **Default:** to use the `default` authentication mechanism, attach the Service Account created above to the Agentless Gateway hosted on a Google Cloud Compute Engine virtual machine instance. For more information on how to attach a Service Account to a GCP Compute Engine virtual machine, see Change the attached service account.

The Service Account should be granted the following roles:

- Pub/Sub Subscriber
- Pub/Sub Viewer
- Viewer

> **Note:**

> It is recommended to keep both the Service Account and the Agentless Gateway Google Cloud VM instance within the same Google Cloud Project.

Google Cloud Log Routing Overview: Log Router Sinks, Pub/Sub Topics and Subscriptions

Log Router Sinks control how Logging routes logs in Google Cloud. Using sinks, you can route audit logs to one or more supported destinations. Please follow the instructions in Route logs to supported destinations. The following summarizes the steps to be completed:

1. Create a Log Router sink that sends the audit logs to a new or existing Pub/Sub Topic as the destination service.
   a. In the "Choose logs to include in sink" panel, filter for your data source’s audit logs. Follow the below examples to create a filter for your data source configuration.
2. Create a Pub/Sub Subscription that receives messages from the Pub/Sub Topic. The Agentless Gateway service will pull the audit logs from this subscription.

The gcloud command line tool - linked to a project owner service account - may also be used to create the Log Router Sink and export the audit logs to a Pub/Sub Topic.

Firestore supports retrieving audit logs in both Native mode and Datastore mode. For more information on these modes, see Choosing between Native mode and Datastore mode.

Routing logs from a specific Firestore database to a Pub/Sub Topic (One-to-One)

The filter can also be customized to include logs from specific Firestore databases. For example, to forward logs from one specific Firestore database in the project, use:

```
protoPayload.resourceName: "projects/my-project-id/databases/my-firestore-database" AND (prc
```

Routing logs from all Firestore databases to a Pub/Sub Topic (Many-to-One)

Use the following filter when creating a Log Router Sink to forward audit logs from all Firestore databases (both Native and Datastore modes) within the project:

```
protoPayload.serviceName="firestore.googleapis.com" OR
protoPayload.serviceName="datastore.googleapis.com"
```

This routes all Firestore audit logs to the specified Pub/Sub subscription.

> **Note:**
>
> It is recommended to have one Agentless Gateway pull from each given Pub/Sub subscription. Multiple

> Agentless Gateways pulling from the same Pub/Sub subscription may result in the ingestion of partial data.

Network Prerequisites

Below is an overview of the necessary requirements to enable the Agentless Gateway to communicate with data sources and other components across the network. Please ensure these are completed by a Network Administrator.

Agentless Gateway Firewall Settings

DSF reads audit events from Google's Pub/Sub service running on the Google cloud. Ensure the Agentless Gateway host is able to access the Pub/Sub subscription. Note that:

- The Agentless Gateway host must be able to reach the Google Cloud Pub/Sub API endpoint `pubsub.googleapis.com`. If the gateway is configured with dynamic subscribers (a non-default setting), it must also reach `monitoring.googleapis.com`.
- The firewall needs to allow traffic from the Agentless Gateway on several outbound ports to enable communication with the Pub/Sub subscription:
  - Ports 80 and 443: for HTTP and HTTPS requests to the Pub/Sub subscription.

✔STEP 2: Enabling Audit on the Data Source

Once the required permissions and information have been obtained, please complete the following steps to enable audit on the data source.

Enable Audit Logging

The Cloud Firestore auditing facility provides three distinct types of messages:

- System Event audit logs: these messages identify automated Google Cloud actions that modify the configuration of resources.
- Admin Activity audit logs: these messages include "admin write" operations that write metadata or configuration information.
- Data Access audit logs: these messages include "admin read" operations that read metadata or configuration information. They also include "data read" and "data write" operations that read or write user provided data.

By default, **System Event** and **Admin Activity** logs are enabled. **Data Access** logs are disabled by default.

To enable Data Access audit logs for Firestore:

1. From the Google Cloud console, navigate to **IAM & Admin** > **Audit Logs**.
2. In the **Data Access audit logs configuration** table, select **Firestore/Datastore API service** under the

Service column.

3. On the right, under the **Log Types** tab, enable the permissions:
   - `ADMIN_READ`
   - `DATA_READ`
   - `DATA_WRITE`
4. Click **Save**.

For more information on audit logging for Firestore, see Firestore audit logging information.

✔STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

GCP Firestore Asset Specifications

GCP Pub Sub Asset Specifications

GCP Asset Specifications

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.

2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.

3. In the Data Source Type dropdown menu, select a data source.

4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.

5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click Advanced.

6. Under Monitoring, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.

7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the Adding Assets via Unified Settings Console (USC) documentation.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

1. From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specification documentation as a guide for filling in the necessary field values.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Management** > **Audit Actions** > **Connect Gateway** to start collecting audit data.

For more information, please see the Adding Assets via the Import Assets Page documentation.

> **Note:**
>
> Asset template name for Cloud Firestore: ***GOOGLECLOUD_FIRESTORE_template.xlsx***
>
> - If you're forwarding audit logs from multiple Firestore databases to a single Pub/Sub topic, you only need to onboard one asset. The `asset_id` field in the spreadsheet template is a placeholder. The actual `asset_id` in the audit logs reflect the originating database.

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see Using DSF Open APIs.

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via https://supportportal.thalesgroup.com/csm.

On the Agentless Gateway host(s), please review the following:

DSF Hub versions 15.3 and newer :

- Gateway log file: $JSONAR_LOGDIR/gateway/syslog/firestore_pubsub.log
- Gateway pull log file: `$JSONAR_LOGDIR/gateway/pull/google-pubsub/gateway-pull.log`
- Run the following command to verify that the required services are active:

```
systemctl status -l gateway-pull@google-pubsub.service
systemctl status -l sonarrsyslog
```

DSF Hub versions 15.1 to 15.2 :

- Gateway log file: $JSONAR_LOGDIR/gateway/syslog/firestore_pubsub.log
- SonarGD log file: `$JSONAR_LOGDIR/sonargd/sonargd.log`
- Run the following command to verify the status of the services:

```
systemctl status -l sonargd
systemctl status -l sonarrsyslog
```

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via https://supportportal.thalesgroup.com/csm. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

- Using DSF Open APIs: A guide to onboarding and managing assets via DSF Open APIs.
- Adding assets via Unified Settings Console (USC): Instructions to add and edit data source assets via the USC.
- Adding Assets via the Import Assets Page: Instructions to add data source assets via the Asset Dashboard.
- Using the Data Security Fabric Portal: An overview of the Data Security Fabric (DSF) portal.
- Data Security Coverage Tool: Systems and version compatibility with DSF Hub.

Document generated by Confluence on Jun 19, 2026 01:21

Atlassian

Spanner Onboarding Steps

This topic reviews the steps required to audit activity on Cloud Spanner databases and send that information to DSF Hub for auditing, analysis, detecting and preventing security events.

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The following main topics are included:

- An overview of the available scope for the data source, given its native audit data.
- A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.
- Instructions on how to enable audit on the data source and collect data using DSF Hub.
- Initial troubleshooting steps and technical support information.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps. Please click on each of the steps below to display the content.

STEP 1: Onboarding Prerequisites

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

Permissions for Google Cloud Account Principal

Grant the following IAM roles to a cloud account principal:

- **Logging Admin** and **Project IAM Admin** roles: for enabling Data Access logs, viewing logs and creating a Log Router Sink.
- **Pub/Sub Admin** role: for creating a Pub/Sub Topic and Subscription.
- **Service Account Admin** and optionally **Service Account Key Admin** roles: for creating a service account and generating keys (if using "service_account" auth mechanism). See Permissions and roles for more information on the list of individual permissions contained in each predefined role.

Creating a Google Cloud Service Account

DSF supports the following two authentication mechanisms for accessing Google Cloud resources. Note that both authentication mechanisms make use of a Service Account. The process for creating a Service Account is described in Creating and managing service accounts.

- **Service Account:** to use the `service_account` authentication mechanism, a JSON-formatted key file needs to be generated holding the credentials that the Agentless Gateway will use to access audit logs from the Pub/Sub subscription. Ensure that the key file is copied to the Agentless Gateway host and it is owned by the `sonarw` user. For more information on how to create a service account key, see Creating and managing service account keys.

- **Default:** to use the `default` authentication mechanism, attach the Service Account created above to the Agentless Gateway hosted on a Google Cloud Compute Engine virtual machine instance. For more information on how to attach a Service Account to a GCP Compute Engine virtual machine, see Change the attached service account.

The Service Account should be granted the following roles:

- Pub/Sub Subscriber
- Pub/Sub Viewer
- Viewer

> **Note:**
>
> It is recommended to keep both the Service Account and the Agentless Gateway Google Cloud VM instance within the same Google Cloud Project.

Google Cloud Log Routing Overview: Log Router Sinks, Pub/Sub Topics and Subscriptions

Log Router Sinks control how Logging routes logs in Google Cloud. Using sinks, you can route audit logs to one or more supported destinations. Please follow the instructions in Route logs to supported destinations. The following summarizes the steps to be completed:

1. Create a Log Router sink that sends the audit logs to a new or existing Pub/Sub Topic as the destination service.
   a. In the "Choose logs to include in sink" panel, filter for your data source’s audit logs. Follow the below examples to create a filter for your data source configuration.
2. Create a Pub/Sub Subscription that receives messages from the Pub/Sub Topic. The Agentless Gateway service will pull the audit logs from this subscription.

The gcloud command line tool - linked to a project owner service account - may also be used to create the Log Router Sink and export the audit logs to a Pub/Sub Topic.

Routing logs of all spanner instances to a Pub/Sub Topic

The below filter should be used while creating the Log Router Sink. This will route the audit logs of all spanner instances in a project to the subscription.

```
resource.type="spanner_instance"
```

Routing logs of specific spanner instances to a Pub/Sub Topic

The filter can also be customized to include logs of specific spanner instances. For example, to send the logs of two spanner instances in a project to the pub/sub use:

```
resource.type="spanner_instance" AND (resource.labels.instance_id="<spanner-instance1-id>" C
```

Creating a Pub/Sub Subscription

Create a Pub/Sub subscription by following the instructions at Create pull subscriptions. For more information on Pub/Sub Topics and Subscriptions, see Overview of the Pub/Sub service and Create a topic.

> **Note:**
>
> - It is recommended to have one Agentless Gateway pull from a given Pub/Sub subscriptions. Multiple Agentless Gateways pulling from the same Pub/Sub subscription may result in the ingestion of partial data.
> - A single Pub/Sub subscription can store the audit logs of multiple instances of the same Server Type. However, the audit logs of different Server Types cannot be sent to the same Pub/Sub subscription.
> - If several instances of the same Server Type write audit data to a single Pub/Sub subscription, connecting any one of the data source assets or the Pub/Sub subscription asset will pull all the audit data from the subscription. The Pub/Sub asset will stay connected to the gateway and will pull all the audit data until all the data source assets using it are disconnected.

Network Prerequisites

Below is an overview of the necessary requirements to enable the Agentless Gateway to communicate with data sources and other components across the network. Please ensure these are completed by a Network Administrator.

Agentless Gateway Firewall Settings

DSF reads audit events from Google's Pub/Sub service running on the Google cloud. Ensure the Agentless Gateway host is able to access the Pub/Sub subscription. Note that:

- The Agentless Gateway host must be able to reach the Google Cloud Pub/Sub API endpoint `pubsub.googleapis.com`. If the gateway is configured with dynamic subscribers (a non-default setting), it must also reach `monitoring.googleapis.com`.
- The firewall needs to allow traffic from the Agentless Gateway on several outbound ports to enable communication with the Pub/Sub subscription:

- Ports 80 and 443: for HTTP and HTTPS requests to the Pub/Sub subscription.

✔STEP 2: Enabling Audit on the Data Source

Once the required permissions and information have been obtained, please complete the following steps to enable audit on the data source.

Enable Audit Logging

The Cloud Spanner auditing facility provides three distinct types of messages:

- System Event audit logs: Identifies automated Google Cloud actions that modify the configuration of resources.
- Admin Activity audit logs: Includes "admin write" operations that write metadata or configuration information.
- Data Access audit logs: Includes "admin read" operations that read metadata or configuration information. Also includes "data read" and "data write" operations that read or write user provided data.

System Event and Admin Activity audit logs are enabled by default. Data Access audit logs are disabled by default. To enable them, execute the following steps:

1. From the Google Cloud console, navigate to IAM & Admin > Audit Logs.
2. In the Data Access audit logs configuration table, select Cloud Spanner API service in the Service column.
3. In the Log Types tab that appears on the right, enable “Admin Read”, “Data Read”, and “Data Write”. For more information, see Enable Data Access audit logs.
4. Click Save.

For more information on Cloud Spanner auditing, please see Spanner audit logging.

✔STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

GCP Spanner Asset Specifications
GCP Pub Sub Asset Specifications

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.

2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.

3. In the Data Source Type dropdown menu, select a data source.

4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.

5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click Advanced.

6. Under Monitoring, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.

7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the Adding Assets via Unified Settings Console (USC) documentation.

> **Note:**
>
> To monitor slow queries, enter an integer value in the **Duration Threshold** field to specify the execution time in milliseconds after which a query is flagged as slow.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

1. From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specification documentation as a guide for filling in the necessary field values.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Management** > **Audit Actions** > **Connect Gateway** to start collecting audit data.

For more information, please see the Adding Assets via the Import Assets Page documentation.

Note:

Asset template spreadsheet name for Cloud Spanner: **GOOGLECLOUD_SPANNER_template.xlsx**

- If you are sending audit logs from multiple Spanner instances to a single Pub/Sub, you may connect or disconnect all instances to or from the Agentless Gateway simultaneously by filtering for the assets in the Assets Dashboard, clicking on the "Management" button displayed above the assets → Audit Actions → Connect/Disconnect Gateway.
- The audit logs will be enriched using the asset document of the Spanner instance. In the case in which a Spanner instance cannot be found, they will be enriched with values from the Pub/Sub asset instead.
- To monitor slow queries, add the **duration_threshold** column to the SPANNER asset row and set an integer value for the query execution time in milliseconds. Queries that exceed this threshold will be flagged as slow.

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see Using DSF Open APIs.

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via https://supportportal.thalesgroup.com/csm.

On the Agentless Gateway host(s) please review the following:

DSF Hub versions 15.3 and newer :

- Gateway log file: $JSONAR_LOGDIR/gateway/syslog/spanner_pubsub.log
- Gateway pull log file: `$JSONAR_LOGDIR/gateway/pull/google-pubsub/gateway-pull.log`

- Run the following command to verify that the required services are active:

```
systemctl status -l gateway-pull@google-pubsub.service
systemctl status -l sonarrsyslog
```

DSF Hub versions 4.11 to 15.2:

- Gateway log file: $JSONAR_LOGDIR/gateway/syslog/spanner_pubsub.log
- SonarGD log file: $JSONAR_LOGDIR/sonargd/sonargd.log

- Run the following command to verify the status of the services:

```
systemctl status -l sonargd
```

```
systemctl status -l sonarrsyslog
```

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via https://supportportal.thalesgroup.com/csm. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

- Using DSF Open APIs: A guide to onboarding and managing assets via DSF Open APIs.
- Adding assets via Unified Settings Console (USC): Instructions to add and edit data source assets via the USC.
- Adding Assets via the Import Assets Page: Instructions to add data source assets via the Asset Dashboard.
- Using the Data Security Fabric Portal: An overview of the Data Security Fabric (DSF) portal.
- Data Security Coverage Tool: Systems and version compatibility with DSF Hub.

Document generated by Confluence on Jun 19, 2026 01:21

Atlassian